

“The Essentials of Computer Organization and Architecture”

Unit 5 – Exercises and Solutions

EAT1 A computational system with a **3-level** memory hierarchy has average access times of 20 ns (level 1), 200 ns (level 2) and 2 ms (level 3). For level 1 and 2, the average hit rates are 90% and 95%.
a) Compute the effective (average) access time of this memory hierarchy.

EAT1 A computational system with a **3-level** memory hierarchy has average access times of 20 ns (level 1), 200 ns (level 2) and 2 ms (level 3). For level 1 and 2, the average hit rates are 90% and 95%.
a) Compute the effective (average) access time of this memory hierarchy.

Solution a):

- $H1 = 0,90$; $H2 = 0,95$
- $T1 = 20 \text{ ns}$; $T2 = 200 \text{ ns}$; $T3 = 2 \text{ ms} = 2 \times 10^6 \text{ ns}$

- **overlapped access:**

$$\begin{aligned} EAT_3 &= H1 \times T1 + (1-H1) \times [H2 \times T2 + (1-H2) \times T3] \\ &= 0,9 \times 20 + (1-0,9) \times [0,95 \times 200 + (1-0,95) \times 2 \times 10^6] \\ &= 18 + 0,1 \times [190 + 100000] \\ &= 18 + 10019 = 10037 \text{ ns} = 10,037 \text{ us} = 0,010037 \text{ ms} \end{aligned}$$

- **sequential access:**

$$\begin{aligned} EAT_3 &= H1 \times T1 + (1-H1) \times [H2 \times (T1+T2) + (1-H2) \times (T1+T2+T3)] \\ &= 0,9 \times 20 + (1-0,9) \times [0,95 \times (20+200) + (1-0,95) \times (20+200+2 \times 10^6)] \\ &= 18 + 0,1 \times [209 + 100011] \\ &= 18 + 10022 = 10040 \text{ ns} = 10,040 \text{ us} = 0,010040 \text{ ms} \end{aligned}$$

EAT1 A computational system with a **3-level** memory hierarchy has average access times of 20 ns (level 1), 200 ns (level 2) and 2 ms (level 3). For level 1 and 2, the average hit rates are 90% and 95%.
b) Compute the probability of having to access levels 2 and 3 (separately).

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Solution b):

- $H1 = 0,90$; $H2 = 0,95$

- **overlapped access:**

$$EAT_3 = H1 \times T1 + (1-H1) \times [H2 \times T2 + (1-H2) \times T3]$$

- **sequential access:**

$$EAT_3 = H1 \times T1 + (1-H1) \times [H2 \times (T1+T2) + (1-H2) \times (T1+T2+T3)]$$

$$P1 = H1 = 0,90$$

(probability of having to access level 1 only)

$$P2 = (1-H1) \times H2 = (1-0,9) \times 0,95 = 0,095$$

(probability of having to access level 2)

$$P3 = (1-H1) \times (1-H2) = (1-0,9) \times (1-0,95) = 0,005$$

(probability of having to access level 3)

$$\text{verification: } P1+P2+P3 = 0,9 + 0,095 + 0,005 = 1$$

EAT2 A system with a **4-level** memory hierarchy has average access times of 1 ns (level 1), 10 ns (level 2), 100 ns (level 3) and 1 μ s (level 4). For the first 3 levels, the average hit rates are 90%, 94% and 97%. Compute the average access time of this hierarchy.

EAT2 A system with a **4-level** memory hierarchy has average access times of 1 ns (level 1), 10 ns (level 2), 100 ns (level 3) and 1 us (level 4). For the first 3 levels, the average hit rates are 90%, 94% and 97%. Compute the average access time of this hierarchy.

Solution:

- $H1 = 0,90$; $H2 = 0,94$; $H3 = 0,97$;
- $T1 = 1 \text{ ns}$; $T2 = 10 \text{ ns}$; $T3 = 100 \text{ ns}$; $T4 = 1 \text{ us} = 1 \times 10^3 = 1000 \text{ ns}$

- **overlapped access:**

$$\begin{aligned} EAT_4 &= H1 \times T1 + (1-H1) \times [H2 \times T2 + (1-H2) \times [H3 \times T3 + (1-H3) \times T4]] \\ &= 0,9 \times 1 + (1-0,9) \times [0,94 \times 10 + (1-0,94) \times [0,97 \times 100 + (1-0,97) \times 1000]] \\ &= 2,6 \text{ ns} \end{aligned}$$

- **sequential access:**

$$\begin{aligned} EAT_4 &= H1 \times T1 + (1-H1) \times [H2 \times (T1+T2) + \\ &\quad (1-H2) \times [H3 \times (T1+T2+T3) + (1-H3) \times (T1+T2+T3+T4)]] \\ &= 0,9 \times 1 + (1-0,9) \times [0,94 \times (1+10) + \\ &\quad (1-0,94) \times [0,97 \times (1+10+100) + (1-0,97) \times (1+10+100+1000)]] \\ &= 2,8 \text{ ns} \end{aligned}$$

EAT3 a) Deduce the EAT expression for a **5-level** memory hierarchy. b) Apply it reusing the times and probabilities of exercise EAT2, and considering a hit-rate in level 4 of 99% and access time of 1 ms in level 5.

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Solution:

a) In the 4-level EAT expression, we must adapt the **terminal** component:

- **overlapped access:**

$$EAT_5 = H1 \times T1 + (1-H1) \times [H2 \times T2 + (1-H2) \times [H3 \times T3 + (1-H3) \times (*)]]$$

$$EAT_5 = H1 \times T1 + (1-H1) \times [H2 \times T2 + (1-H2) \times [H3 \times T3 + (1-H3) \times [H4 \times T4 + (1-H4) \times T5]]]$$

- **sequential access:**

$$EAT_5 = H1 \times T1 + (1-H1) \times [H2 \times (T1+T2) + (1-H2) \times [H3 \times (T1+T2+T3) + (1-H3) \times (*)]]$$

$$EAT_5 = H1 \times T1 + (1-H1) \times [H2 \times (T1+T2) + (1-H2) \times [H3 \times (T1+T2+T3) + (1-H3) \times [H4 \times (T1+T2+T3+T4) + (1-H4) \times (T1+T2+T3+T4+T5)]]]$$

EAT3 a) Deduce the EAT expression for a **5-level** memory hierarchy. b) Apply it reusing the times and probabilities of exercise EAT2, and considering a hit-rate in level 4 of 99% and access time of 1 ms in level 5.

Solution:

b)

$H1 = 0,90$; $H2 = 0,94$; $H3 = 0,97$; **$H4 = 0,99$**

$T1 = 1 \text{ ns}$; $T2 = 10 \text{ ns}$; $T3 = 100 \text{ ns}$; $T4 = 1000 \text{ ns}$; **$T5 = 1\text{ms} = 1000000\text{ns}$**

- **overlapped access:**

$$EAT_5 = H1 \times T1 + (1-H1) \times [H2 \times T2 + (1-H2) \times [H3 \times T3 + (1-H3) \times [H4 \times T4 + (1-H4) \times T5]]]$$

$$= 0,9 \times 1 + (1-0,9) \times [0,94 \times 10 + (1-0,94) \times [0,97 \times 100 + (1-0,97) \times [0,99 \times 1000 + (1-0,99) \times 1000000]]] = 4,4 \text{ ns}$$

- **sequential access:**

$$EAT_5 = H1 \times T1 + (1-H1) \times [H2 \times (T1+T2) + (1-H2) \times [H3 \times (T1+T2+T3) + (1-H3) \times [H4 \times (T1+T2+T3+T4) + (1-H4) \times (T1+T2+T3+T4+T5)]]]$$

$$= 0,9 \times 1 + (1-0,9) \times [0,94 \times (1+10) + (1-0,94) \times [0,97 \times (1+10+100) + (1-0,97) \times [0,99 \times (1+10+100+1000) + (1-0,99) \times (1+10+100+1000+1000000)]]] = 4,6 \text{ ns}$$

MP1 Consider a 2M x 32 bit memory. Answer the following questions:



- a) What is the storage capacity of this memory (in bytes) ?**
- b) How many bits have the addresses of this memory, if the memory is byte-addressable?**
- c) c) How many bits have the addresses of this memory, if the memory is word-addressable? (note: consider the word-size as 32 bits)**

MP1 Consider a 2M x 32 bit memory. a) What is the storage capacity of this memory (in bytes) ?

Solution:

	8bits	8bits	8bits	8bits
line 0				
line 1				
	...	...	...	...
line 2M-1				

$$\begin{aligned}\text{\#BYTES(memory)} &= \text{\#LINES(memory)} \times \text{\#BYTES (line)} \\ &= 2\text{M} \times (32 / 8) \\ &= 2\text{M} \times 4 \\ &= 8\text{M bytes}\end{aligned}$$

MP1 Consider a 2M x 32 bit memory. b) How many bits have the addresses of this memory, if the memory is byte-addressable?

Solution:

	8bits	8bits	8bits	8bits
line 0				
line 1				
	...	...	...	...
line 2M-1				

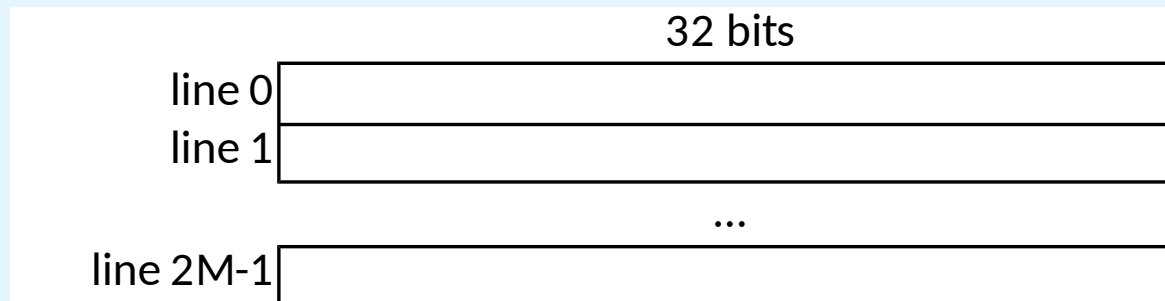
“byte-addressable memory” => CELL == 1 byte == 8bits

$$\begin{aligned}\#CELLS(\text{memory}) &= \#LINES(\text{memory}) \times \#CELLS(\text{line}) \\ &= 2M \times 4 \\ &= 8M \text{ cells}\end{aligned}$$

$$\begin{aligned}\#BITS(\text{address}) &= \log_2(\#CELLS(\text{memory})) = \log_2(8M) \\ &= \log_2(2^3 \times 2^{20}) = \log_2(2^{23}) \\ &= \mathbf{23 \text{ bits}}\end{aligned}$$

MP1 Consider a 2M x 32 bit memory. c) How many bits have the addresses of this memory, if the memory is word-addressable? (note: consider the word-size as 32 bits)

Solution:



“word-addressable memory” $\Rightarrow$ CELL == 32bits == 4bytes

$$\begin{aligned}\#\text{CELLS}(\text{memory}) &= \#\text{LINES}(\text{memory}) \times \#\text{CELLS}(\text{line}) \\ &= 2\text{M} \times 1 \\ &= 2\text{M cells}\end{aligned}$$

$$\begin{aligned}\#\text{BITS}(\text{address}) &= \log_2 (\#\text{CELLS}(\text{memory})) = \log_2 (2\text{M}) \\ &= \log_2 (2^1 \times 2^{20}) = \log_2 (2^{21}) \\ &= \mathbf{21 \text{ bits}}\end{aligned}$$

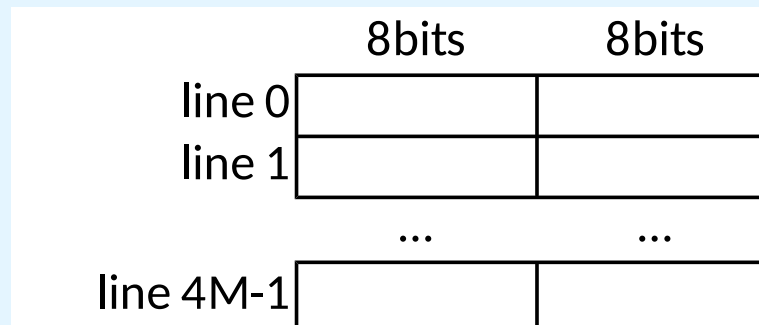
MP2 Consider a 4M x 16 bit memory. Answer the following questions:



- a) What is the storage capacity of this memory (in bytes) ?**
- b) How many bits have the addresses of this memory, if the memory is byte-addressable?**
- c) How many bits have the addresses of this memory, if the memory is word-addressable? (note: consider the word-size as 16 bits)**

MP2 Consider a 4M x 16 bit memory. a) What is the storage capacity of this memory (in bytes) ?

Solution:



$$\begin{aligned}\text{\#BYTES(memory)} &= \text{\#LINES(memory)} \times \text{\#BYTES (line)} \\ &= 4\text{M} \times (16 / 8) \\ &= 4\text{M} \times 2 \\ &= 8\text{M bytes}\end{aligned}$$

MP2 Consider a 4M x 16 bit memory. b) How many bits have the addresses of this memory, if the memory is byte-addressable?

Solution:

	8bits	8bits
line 0		
line 1		
	...	...
line 4M-1		

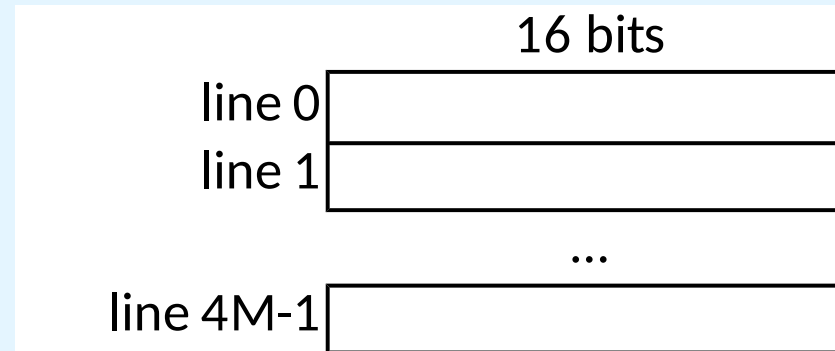
“byte-addressable memory” $\Rightarrow$ CELL == 1 byte == 8bits

$$\begin{aligned}\#CELLS(\text{memory}) &= \#LINES(\text{memory}) \times \#CELLS(\text{line}) \\ &= 4M \times 2 \\ &= 8M \text{ cells}\end{aligned}$$

$$\begin{aligned}\#BITS(\text{address}) &= \log_2(\#CELLS(\text{memory})) = \log_2(8M) \\ &= \log_2(2^3 \times 2^{20}) = \log_2(2^{23}) \\ &= \mathbf{23 \text{ bits}}\end{aligned}$$

MP2 Consider a 4M x 16 bit memory. c) How many bits have the addresses of this memory, if the memory is word-addressable? (note: consider the word-size as 16 bits)

Solution:



“word-addressable memory” $\Rightarrow$ CELL == 16bits == 2bytes

$$\begin{aligned}\#\text{CELLS}(\text{memory}) &= \#\text{LINES}(\text{memory}) \times \#\text{CELLS}(\text{line}) \\ &= 4\text{M} \times 1 \\ &= 4\text{M cells}\end{aligned}$$

$$\begin{aligned}\#\text{BITS}(\text{address}) &= \log_2 (\#\text{CELLS}(\text{memory})) = \log_2 (4\text{M}) \\ &= \log_2 (2^2 \times 2^{20}) = \log_2 (2^{22}) \\ &= \mathbf{22 \text{ bits}}\end{aligned}$$

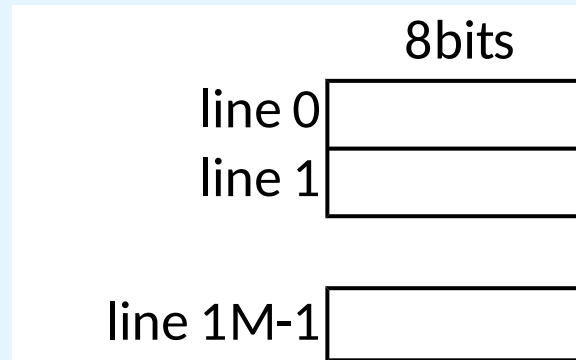
MP3 Consider a 1M x 8 bit memory.

Answer the following questions:

- a) How many bits have the addresses of this memory, if the memory is byte-addressable?**
- b) How many bits have the addresses of this memory, if the memory is word-addressable?**

MP3 Consider a 1M x 8 bit memory. a) How many bits have the addresses of this memory, if the memory is byte-addressable?

Solution:



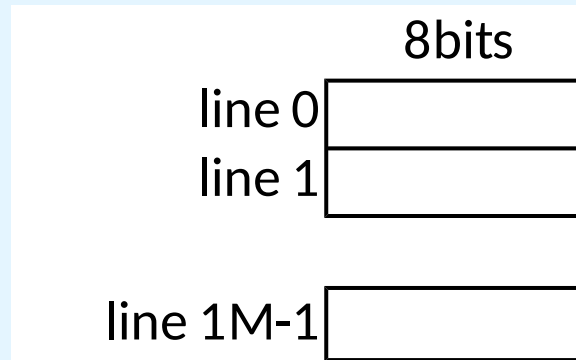
“byte-addressable memory” $\Rightarrow$ CELL == 1 byte == 8bits

$$\begin{aligned}\# \text{CELLS}(\text{memory}) &= \# \text{LINES}(\text{memory}) \times \# \text{CELLS}(\text{line}) \\ &= 1\text{M} \times 1 \\ &= 1\text{M cells}\end{aligned}$$

$$\begin{aligned}\# \text{BITS}(\text{address}) &= \log_2(\# \text{CELLS}(\text{memory})) = \log_2(1\text{M}) \\ &= \log_2(2^0 \times 2^{20}) = \log_2(2^{20}) \\ &= \mathbf{20 \text{ bits}}\end{aligned}$$

MP3 Consider a 1M x 8 bit memory. b) How many bits have the addresses of this memory, if the memory is word-addressable?

Solution:



“word-addressable memory” $\Rightarrow$ CELL == 8bits == 1bytes
(in this scenario, the word is restricted to the minimum of a single byte)

$$\begin{aligned}\#CELLS(\text{memory}) &= \#LINES(\text{memory}) \times \#CELLS(\text{line}) \\ &= 1M \times 1 \\ &= 1M \text{ cells}\end{aligned}$$

$$\begin{aligned}\#BITS(\text{address}) &= \log_2 (\#CELLS(\text{memory})) = \log_2 (1M) \\ &= \log_2 (2^0 \times 2^{20}) = \log_2 (2^{20}) \\ &= \mathbf{20 \text{ bits}}\end{aligned}$$

MP4 Consider a memory of 2^{20} bytes. Answer the following questions:



- a) What is the lowest and highest address if this memory is byte-addressable?**
- b) What is the lowest and highest address if this memory is word-addressable?
(note: consider the word size as 16 bits)**
- c) What is the lowest and highest address if this memory is word-addressable?
(note: consider the word size as 32 bits)**

MP4 Consider a memory of 2^{20} bytes: a) What is the lowest and highest address if this memory is byte-addressable?

Solution:

“byte addressable memory” $\Rightarrow$ #BYTES(cell) = 1 byte

$$\begin{aligned}\text{\#CELLS(memory)} &= \text{\#BYTES(memory)} / \text{\#BYTES(cell)} \\ &= (2^{20}) / 1 \\ &= 2^{20} \text{ cells}\end{aligned}$$

address range = $0 \dots \text{\#CELLS(memory)} - 1 = 0 \dots 2^{20} - 1$

MP4 Consider a memory of 2^{20} bytes: b) What is the lowest and highest address if this memory is word-addressable? (note: consider the word size as 16 bits)

Solution:

“16-bit word addressable memory” $\Rightarrow$ #BYTES(cell) = 2 bytes

$$\begin{aligned}\text{\#CELLS(memory)} &= \text{\#BYTES(memory)} / \text{\#BYTES(cell)} \\ &= (2^{20}) / 2 \\ &= 2^{19} \text{ cells}\end{aligned}$$

$$\text{address range} = 0 \dots \text{\#CELLS(memory)} - 1 = 0 \dots 2^{19} - 1$$

MP4 Consider a memory of 2^{20} bytes: c) What is the lowest and highest address if this memory is word-addressable? (note: consider the word size as 32 bits)

Solution:

“32-bit word addressable memory” $\Rightarrow$ #BYTES(cell) = 4 bytes

$$\begin{aligned}\text{\#CELLS(memory)} &= \text{\#BYTES(memory)} / \text{\#BYTES(cell)} \\ &= (2^{20}) / 4 \\ &= 2^{18} \text{ cells}\end{aligned}$$

$$\text{address range} = 0 \dots \text{\#CELLS(memory)} - 1 = 0 \dots 2^{18} - 1$$

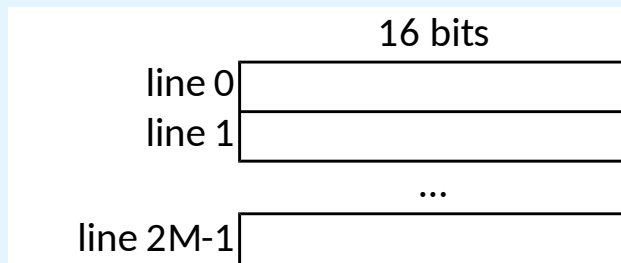
MP5 A $2\text{M} \times 16$ main memory is built using $256\text{K} \times 8$ chips and memory is word-addressable. Provide:

- a) The number of chips necessary per each memory word.**
- b) The total number of modules of the memory.**
- c) The total number of chips of the memory.**
- d) The number of bits of the memory addresses.**
- e) The number of bits used to select the module.**
- f) The number of bits used to define the offset (line) within a module.**
- g) The module and the offset (line) within that module, for the memory address 14, under g1) High-Order and g3) Low-Order interleaving**
- h) The overall storage capacity (in bytes) of the memory.**

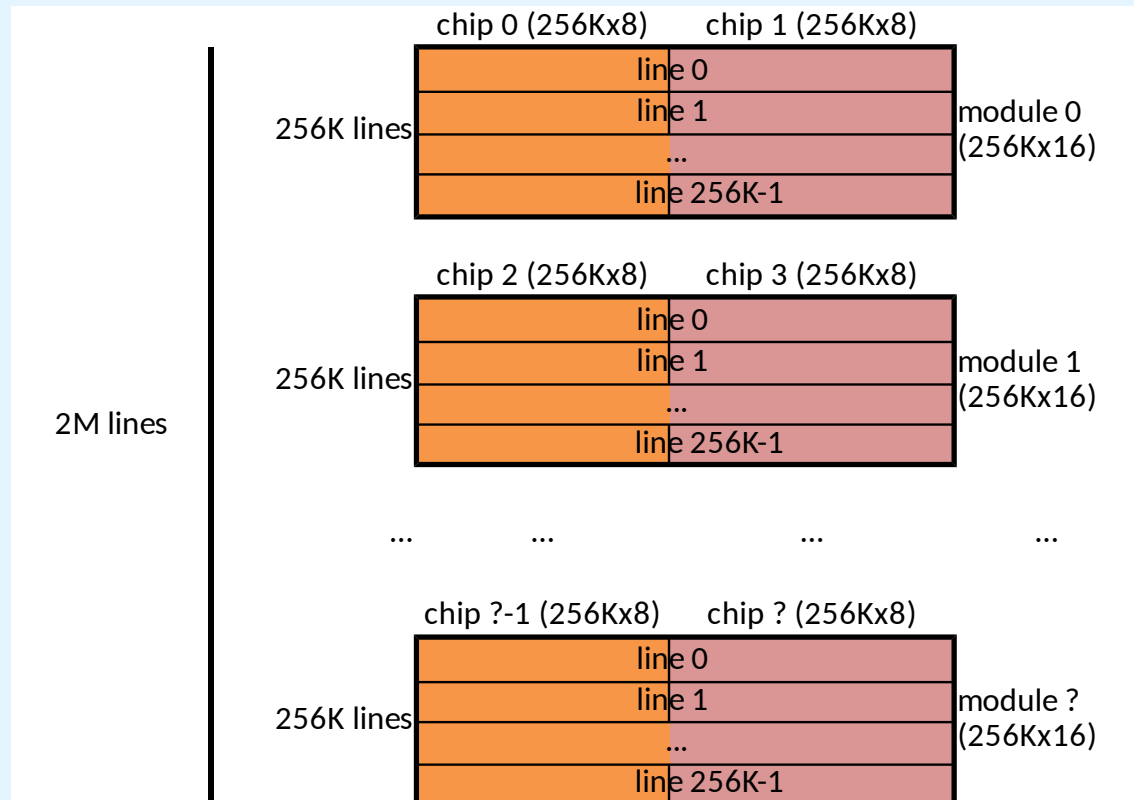
MP5 A 2M x 16 main memory is built using 256K × 8 chips and memory is word-addressable.

Based on the data given, one can build the following representations:

Abstract View of Memory



Physical View of Memory



MP5 A 2M x 16 main memory is built using 256K x 8 chips and memory is word-addressable. a) How many chips are there per memory word?

Solution:

$$\begin{aligned}\text{\textcolor{red}{\#CHIPS(word)}} &= \text{\#BITS(word)} / \text{\#BITS(chip width)} \\ &= \text{\#BITS(memory width)} / \text{\#BITS(chip width)} \\ &= 16 / 8 \\ &= \text{\textcolor{red}{2 chips}}\end{aligned}$$

(therefore, **\text{\textcolor{red}{\#CHIPS(module)}}** is also **\text{\textcolor{red}{2 chips}}**)

(also, a module has the same height of a chip,
that is, **\text{\#LINES(module)} = \text{\#LINES(chip)} = 256K**)

MP5 A 2M x 16 main memory is built using 256K x 8 chips and memory is word-addressable. b) How many RAM modules will this memory have?

Solution:

$$\text{\#MODULES(memory)} = \text{\#LINES(memory)} / \text{\#LINES(module)}$$

$$= 2\text{M} / 256\text{K}$$

$$= 2^{21} / 2^{18}$$

$$= 2^3$$

$$= \mathbf{8 \text{ modules}}$$

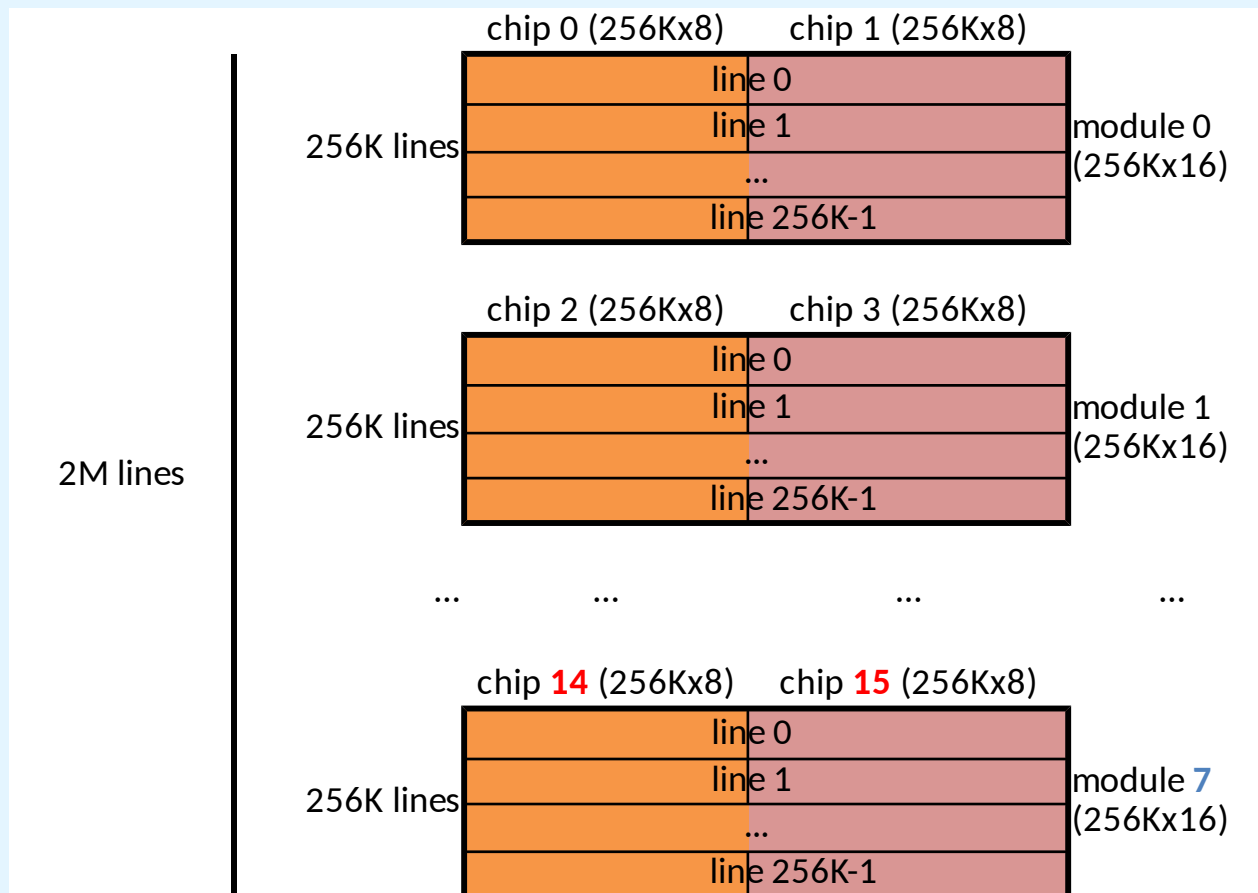
MP5 A 2M x 16 main memory is built using 256K x 8 chips and memory is word-addressable. c) How many chips will this memory have?

Solution:

$$\begin{aligned}\#CHIPS(memory) &= \#MODULES(memory) \times \#CHIPS(module) \\ &= 8 \times 2 \\ &= \mathbf{16 \text{ chips}}\end{aligned}$$

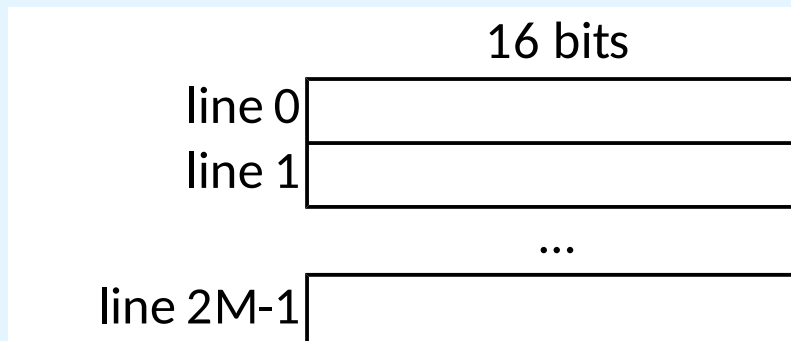
MP5 A 2M x 16 main memory is built using 256K × 8 chips and memory is word-addressable.

Accordingly with the previous calculations, this is the updated Physical View of the Memory:



MP5 A 2M x 16 main memory is built using 256K x 8 chips and memory is word-addressable. d) How many address bits are needed for all of memory?

Solution:



“word-addressable memory” => CELL == 16bits

$$\begin{aligned}\#CELLS(\text{memory}) &= \#LINES(\text{memory}) \times \#CELLS(\text{line}) \\ &= 2M \times 1 \\ &= 2M \text{ cells}\end{aligned}$$

$$\begin{aligned}\#BITS(\text{memory address}) &= \log_2 (\#CELLS(\text{memory})) = \log_2 (2M) \\ &= \log_2 (2^1 \times 2^{20}) = \log_2 (2^{21}) \\ &= \mathbf{21 \text{ bits}}\end{aligned}$$

MP5 A 2M x 16 main memory is built using 256K x 8 chips and memory is word-addressable. e) How many address bits are needed to identify a module ?

Solution:

$$\text{\#BITS(module address)} = \log_2 (\text{\#MODULES(memory)})$$

$$= \log_2 (8)$$

$$= \log_2 (2^3)$$

$$= \mathbf{3 \text{ bits}}$$

MP5 A 2M x 16 main memory is built using 256K x 8 chips and memory is word-addressable. f) How many address bits are needed to identify a module line ?

Solution:

$$\begin{aligned}\text{\textcolor{red}{\#BITS(module line address)}} &= \log_2 (\text{\#LINES(module)}) \\ &= \log_2 (256K) \\ &= \log_2 (2^8 \times 2^{10}) \\ &= \log_2 (2^{18}) \\ &= \text{\textcolor{red}{18 bits}}\end{aligned}$$

$$\begin{aligned}\text{[or } \text{\textcolor{red}{\#BITS(module line address)}} &= \text{\#BITS(memory address)} - \\ &\quad \text{\textcolor{blue}{\#BITS(module address)}} \\ &= 21 - 3 = \text{\textcolor{red}{18 bits}} \text{]}\end{aligned}$$

MP5 A 2M x 16 main memory is built using 256K × 8 chips and memory is word-addressable. g) Where (module and line) would address **14** be located under g1) High-Order Interleaving, g2) Low-order Interleaving ?

Solution:

$$14_{10} = 8+4+2 = 1110_2 = 00000000000000000001110_2 \text{ (with 21 bits)}$$

g1) with High-Order-Interleaving

$$14 = \begin{array}{|c|c|c|c|} \hline & \text{module} & & \text{module line} \\ \hline & 0 & 0 & 0 \\ \hline \end{array} \begin{array}{|c|} \hline 0000000000000000001110 \\ \hline \end{array}$$

thus, **module = 0** and **module line = 14**

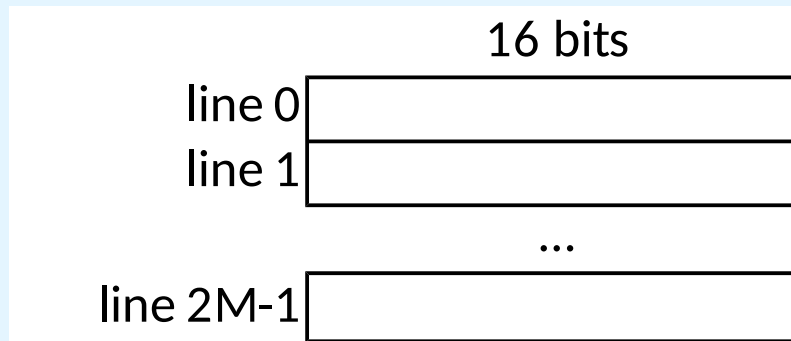
g2) with Low-Order-Interleaving

$$14 = \begin{array}{|c|} \hline 0000000000000000001 \\ \hline \end{array} \begin{array}{|c|c|c|} \hline \text{module line} & & \text{module} \\ \hline 1 & 1 & 0 \\ \hline \end{array}$$

thus, **module = 6** and **module line = 1**

MP5 A 2M x 16 main memory is built using 256K x 8 chips and memory is word-addressable. h) What is the storage capacity of this memory (in bytes) ?

Solution:



$$\begin{aligned}\text{\#BYTES(memory)} &= \text{\#LINES(memory)} \times \text{\#BYTES (line)} \\ &= 2\text{M} \times (16 / 8) \\ &= 2\text{M} \times 2 \\ &= 4\text{M bytes}\end{aligned}$$

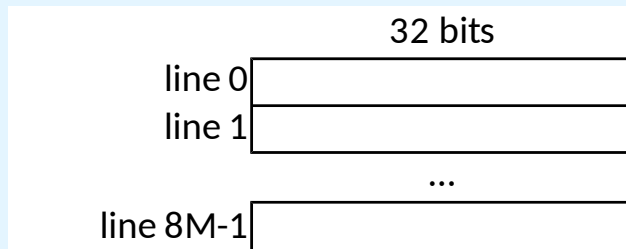
MP6 A 8M x 32 main memory is built using 512K × 16 chips and memory is word-addressable. Provide:

- a) The number of chips necessary per each memory word.**
- b) The total number of modules of the memory.**
- c) The total number of chips of the memory.**
- d) The number of bits of the memory addresses.**
- e) The number of bits used to select the module.**
- f) The number of bits used to define the offset (line) within a module.**
- g) The module and the offset (line) within that module, for the memory address 41, under g1) High-Order and g3) Low-Order interleaving**
- h) The overall storage capacity (in bytes) of the memory.**

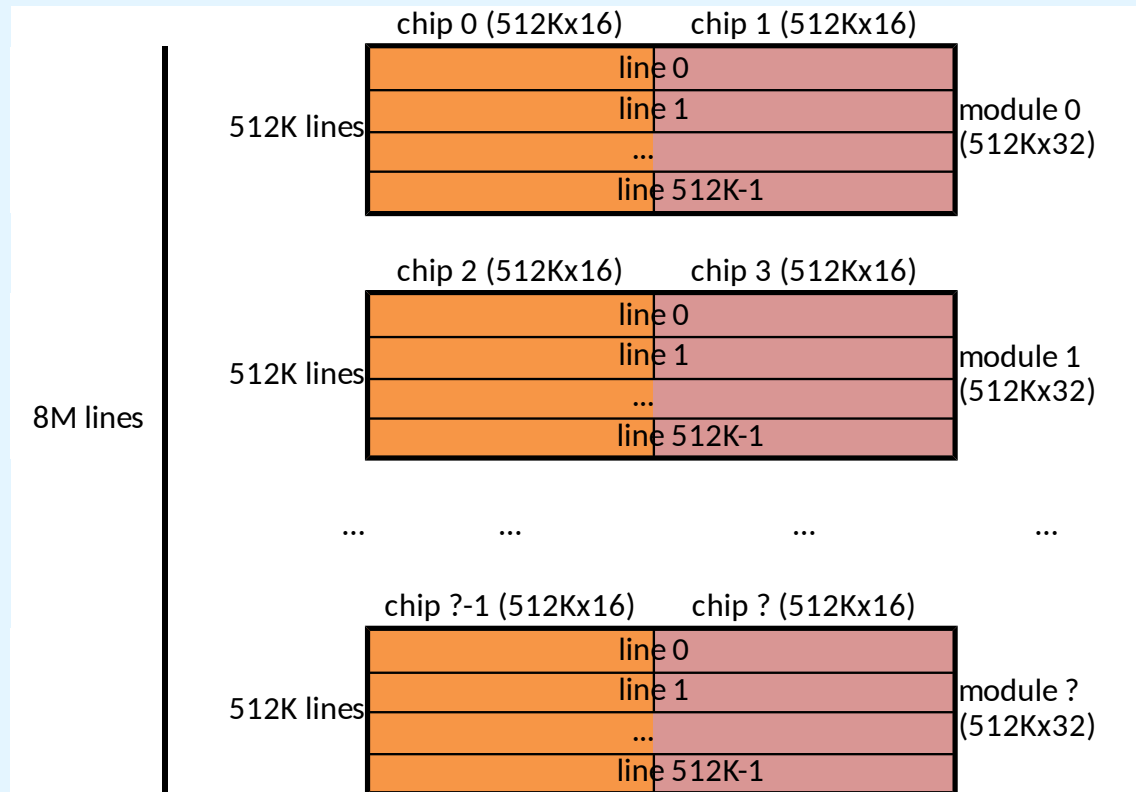
MP6 A 8M x 32 main memory is built using 512K x 16 chips and memory is word-addressable.

Based on the data given, one can build the following representations:

Abstract View of Memory



Physical View of Memory



MP6 A 8M x 32 main memory is built using 512K x 16 chips and memory is word-addressable.
a) How many chips are there per memory word?

Solution:

$$\begin{aligned}\text{\textcolor{red}{\#CHIPS(word)}} &= \text{\#BITS(word)} / \text{\#BITS(chip width)} \\ &= \text{\#BITS(memory width)} / \text{\#BITS(chip width)} \\ &= 32 / 16 \\ &= \text{\textcolor{red}{2 chips}}\end{aligned}$$

(therefore, **\text{\textcolor{red}{\#CHIPS(module)}}** is also **\text{\textcolor{red}{2 chips}}**)

(also, a module has the same height of a chip,
that is, **\text{\#LINES(module)} = \text{\#LINES(chip)} = 512K**)

MP6 A 8M x 32 main memory is built using 512K x 16 chips and memory is word-addressable.
b) How many RAM modules will this memory have?

Solution:

$$\begin{aligned}\text{\#MODULES(memory)} &= \text{\#LINES(memory)} / \text{\#LINES(module)} \\ &= 8\text{M} / 512\text{K} \\ &= 2^{23} / 2^{19} \\ &= 2^4 \\ &= \textbf{16 modules}\end{aligned}$$

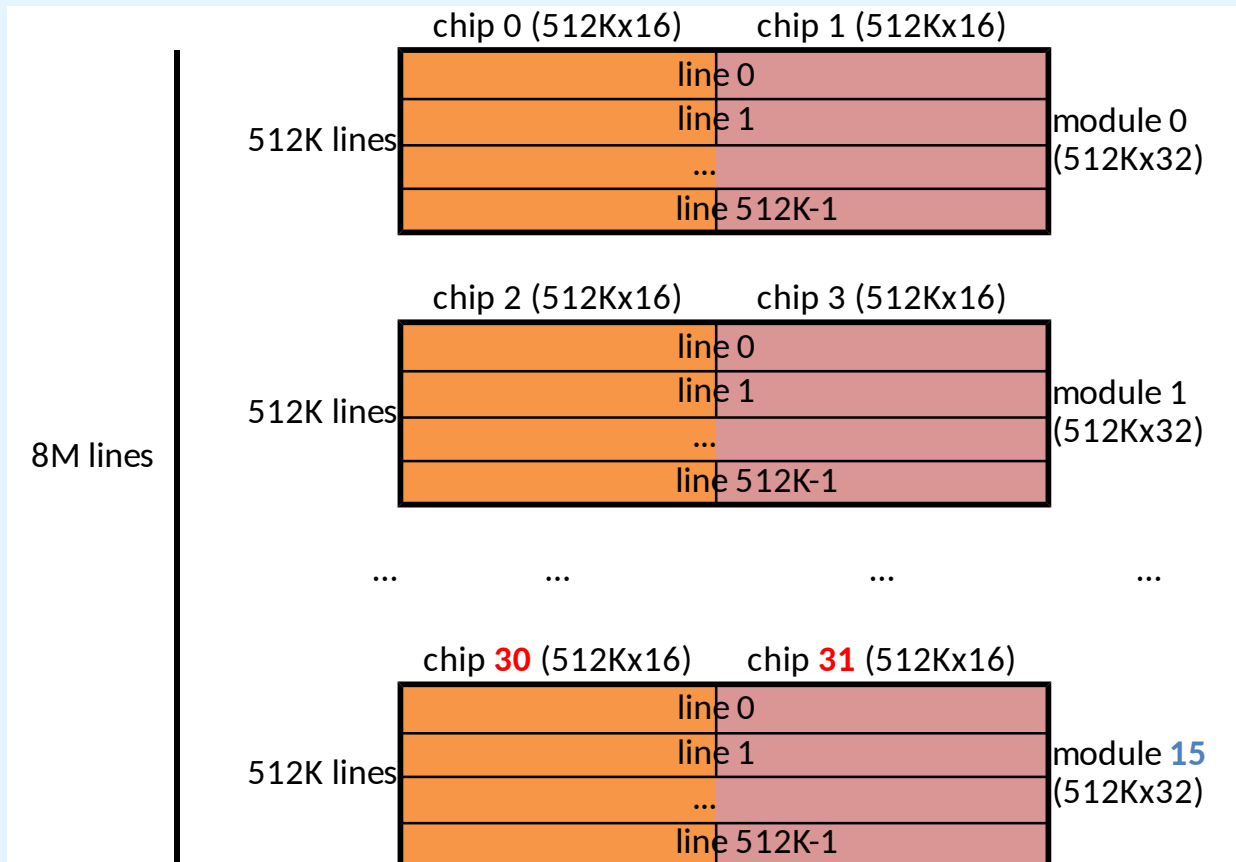
MP6 A 8M x 32 main memory is built using 512K x 16 chips and memory is word-addressable.
c) How many chips will this memory have?

Solution:

$$\begin{aligned}\text{\#CHIPS(memory)} &= \text{\#MODULES(memory)} \times \text{\#CHIPS(module)} \\ &= 16 \times 2 \\ &= \mathbf{32 \text{ chips}}\end{aligned}$$

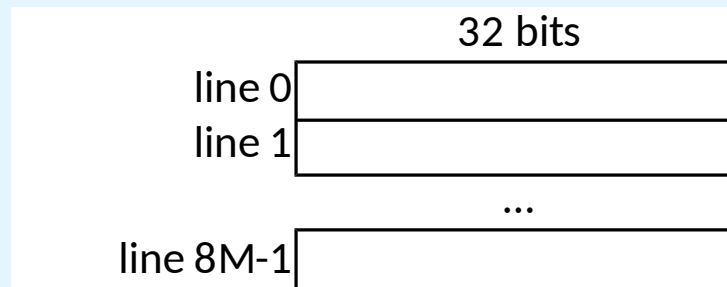
MP6 A 8M x 32 main memory is built using 512K × 16 chips and memory is word-addressable.

Accordingly with the previous calculations, this is the updated Physical View of the Memory:



MP6 A 8M x 32 main memory is built using 512K x 16 chips and memory is word-addressable. d) How many address bits are needed for all of memory?

Solution:



“word-addressable memory” => CELL == 32bits

$$\begin{aligned}\#CELLS(\text{memory}) &= \#LINES(\text{memory}) \times \#CELLS(\text{line}) \\ &= 8M \times 1 \\ &= 8M \text{ cells}\end{aligned}$$

$$\begin{aligned}\#BITS(\text{memory address}) &= \log_2 (\#CELLS(\text{memory})) = \log_2 (2M) \\ &= \log_2 (2^3 \times 2^{20}) = \log_2 (2^{23}) \\ &= \mathbf{23 \text{ bits}}\end{aligned}$$

MP6 A 8M x 32 main memory is built using 512K x 16 chips and memory is word-addressable. e) How many address bits are needed to identify a module ?

Solution:

$$\text{\#BITS(module address)} = \log_2 (\text{\#MODULES(memory)})$$

$$= \log_2 (16)$$

$$= \log_2 (2^4)$$

$$= 4 \text{ bits}$$

MP6 A 8M x 32 main memory is built using 512K x 16 chips and memory is word-addressable. f) How many address bits are needed to identify a module line ?

Solution:

$$\begin{aligned}\text{\#BITS(module line address)} &= \log_2 (\text{\#LINES(module)}) \\ &= \log_2 (512K) \\ &= \log_2 (2^9 \times 2^{10}) \\ &= \log_2 (2^{19}) \\ &= \textbf{19 bits}\end{aligned}$$

$$\begin{aligned}[\text{ or } \text{\#BITS(module line address)} &= \text{\#BITS(memory address)} - \\ &\quad \text{\#BITS(module address)} \\ &= 23 - 4 = \textbf{19 bits}]\end{aligned}$$

MP6 A 8M x 32 main memory is built using 512K x 16 chips and memory is word-addressable. g) Where (module and line) would address 41 be located under g1) High-Order Interleaving, g2) Low-order Interleaving ?

Solution:

$$41_{10} = 32 + 8 + 1 = 101001_2 = \mathbf{0000000000000000000101001}_2 \text{ (with 23 bits)}$$

g1) with High-Order-Interleaving

$$41 = \begin{array}{c|c|c} & \text{module} & \text{module line} \\ \hline & 0 & 0000000000000000000101001 \end{array}$$

thus, **module = 0** and **module line = 41**

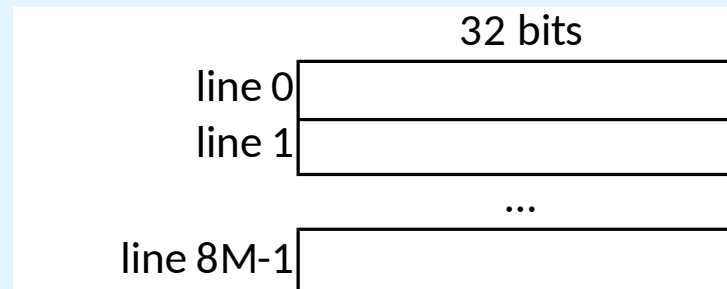
g2) with Low-Order-Interleaving

$$41 = \begin{array}{c|c|c} & \text{module line} & \text{module} \\ \hline & 000000000000000000010 & 1001 \end{array}$$

thus, **module = 9** and **module line = 2**

MP6 A 8M x 32 main memory is built using 512K x 16 chips and memory is word-addressable. h) What is the storage capacity of this memory (in bytes) ?

Solution:



$$\begin{aligned}\text{\#BYTES(memory)} &= \text{\#LINES(memory)} \times \text{\#BYTES (line)} \\ &= 8\text{M} \times (32 / 8) \\ &= 8\text{M} \times 4 \\ &= 32\text{M bytes}\end{aligned}$$

MP7 A memory of 32G x 64 bits is built with 1G x 32 chips and is word-addressable (64 bits word size). Answer the following questions.

a) Provide the number of chips necessary per each memory word:

{?} chips

b) Provide the total number of modules of the memory:

{?} modules

c) Provide the total number of chips of the memory:

{?} chips

d) Provide the number of bits of the memory addresses:

{?} bits

e) Provide the number of bits used to select the module:

{?} bits

f) Provide the number of bits used to define the offset (line) within a module:

{?} bits

MP7 A memory of 32G x 64 bits is built with 1G x 32 chips and is word-addressable (64 bits word size). Answer the following questions.

Solution:

a) Provide the number of chips necessary per each memory word:
{2} chips

b) Provide the total number of modules of the memory:
{32} modules

c) Provide the total number of chips of the memory:
{64} chips

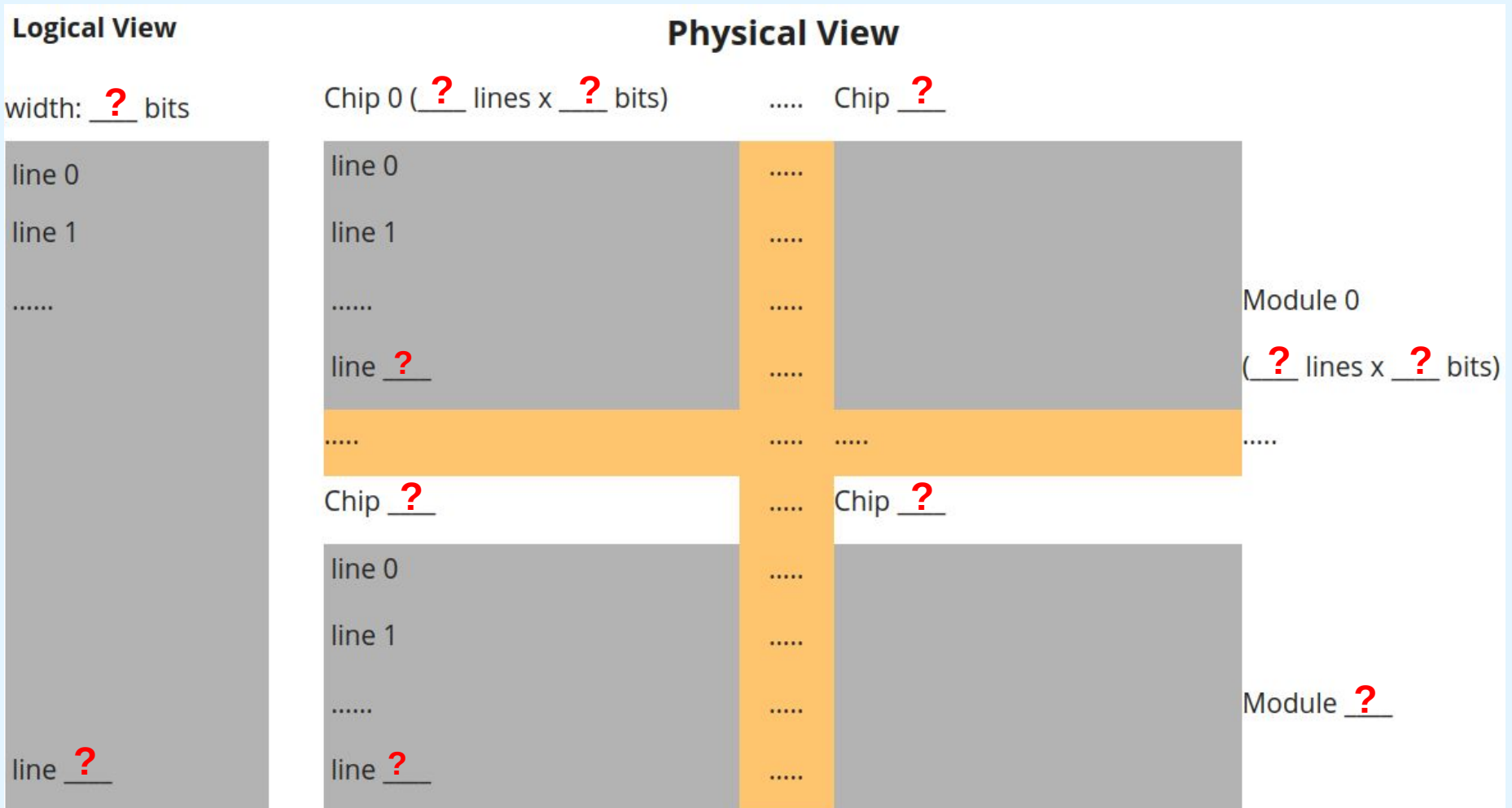
d) Provide the number of bits of the memory addresses:
{35} bits

e) Provide the number of bits used to select the module:
{5} bits

f) Provide the number of bits used to define the offset (line) within a module:
{30} bits

MP7 A memory of 32G x 64 bits is built with 1G x 32 chips and is word-addressable (64 bits word size). Answer the following questions.

g) Complete the representation bellow of the logical & physical view of memory:



built with
64 bits word

Logical View

width: 64 bits

line 0

line 1

.....

line 32G-1

Physical View

Chip 0 (1G lines x 32 bits) Chip 1

line 0

line 1

.....

line 1G-1

.....

Chip 62 Chip 63

line 0

line 1

.....

line 1G-1

.....

Module 0 (1G lines x 64 bits)

Module 31

MP7 A memory of 32G x 64 bits is built with 1G x 32 chips and is word-addressable (64 bits word size). Answer the following questions.

h) Find and present the module and the offset (line) within that module, for the memory address 3333, considering the following alternatives for the mapping of a memory address into its physical components:

h1) High-Order Interleaving:

module: {?}

offset: {?}

h2) Low-Order Interleaving:

module: {?}

offset: {?}

MP7 A memory of 32G x 64 bits is built with 1G x 32 chips and is word-addressable (64 bits word size). Answer the following questions.

h) Find and present the module and the offset (line) within that module, for the memory address 3333, considering the following alternatives for the mapping of a memory address into its physical components:

Solution:

h1) High-Order Interleaving:

module: {0}
offset: {3333}

h2) Low-Order Interleaving:

module: {5}
offset: {104}

MP7 A memory of 32G x 64 bits is built with 1G x 32 chips and is word-addressable (64 bits word size). Answer the following questions.

i) Provide the overall storage capacity (in bytes) of the memory.

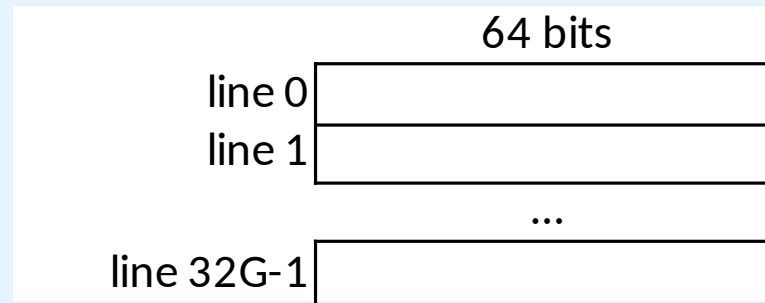
{?} bytes

MP7 A memory of 32G x 64 bits is built with 1G x 32 chips and is word-addressable (64 bits word size). Answer the following questions.

i) Provide the overall storage capacity (in bytes) of the memory.

Solution:

{**256G**} bytes



$$\begin{aligned}\text{\#BYTES(memory)} &= \text{\#LINES(memory)} \times \text{\#BYTES (line)} \\ &= 32\text{G} \times (64 / 8) \\ &= 32\text{G} \times 8 \\ &= 256\text{G bytes}\end{aligned}$$

MP8 A memory of 16G x 32 bits is built with 256M x 8 chips and is word-addressable (32 bits word size). Answer the following questions.

a) Provide the number of chips necessary per each memory word:

{?} chips

b) Provide the total number of modules of the memory:

{?} modules

c) Provide the total number of chips of the memory:

{?} chips

d) Provide the number of bits of the memory addresses:

{?} bits

e) Provide the number of bits used to select the module:

{?} bits

f) Provide the number of bits used to define the offset (line) within a module:

{?} bits

MP8 A memory of 16G x 32 bits is built with 256M x 8 chips and is word-addressable (32 bits word size). Answer the following questions.

Solution:

a) Provide the number of chips necessary per each memory word:
{4} chips

b) Provide the total number of modules of the memory:
{64} modules

c) Provide the total number of chips of the memory:
{256} chips

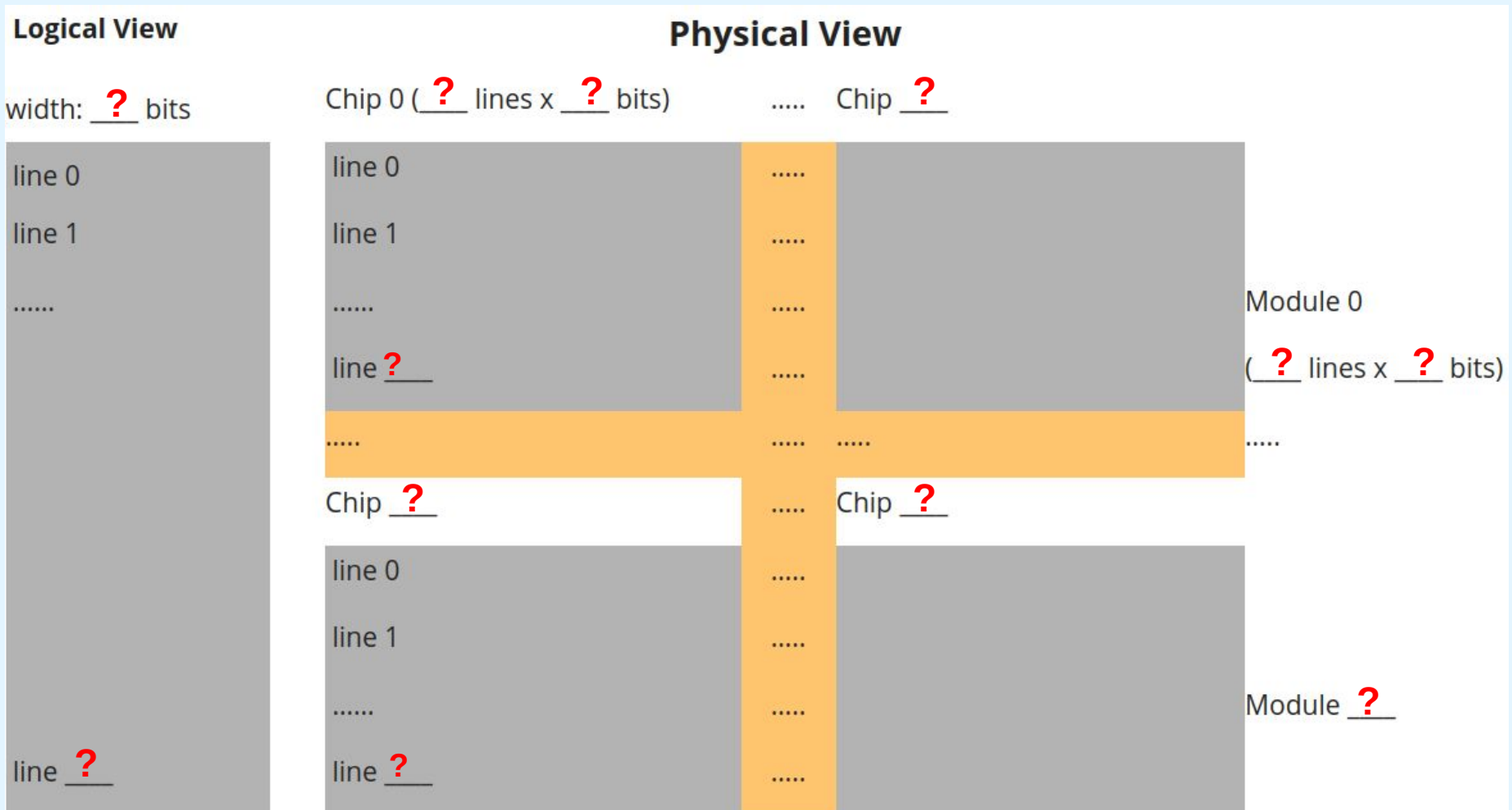
d) Provide the number of bits of the memory addresses:
{34} bits

e) Provide the number of bits used to select the module:
{6} bits

f) Provide the number of bits used to define the offset (line) within a module:
{28} bits

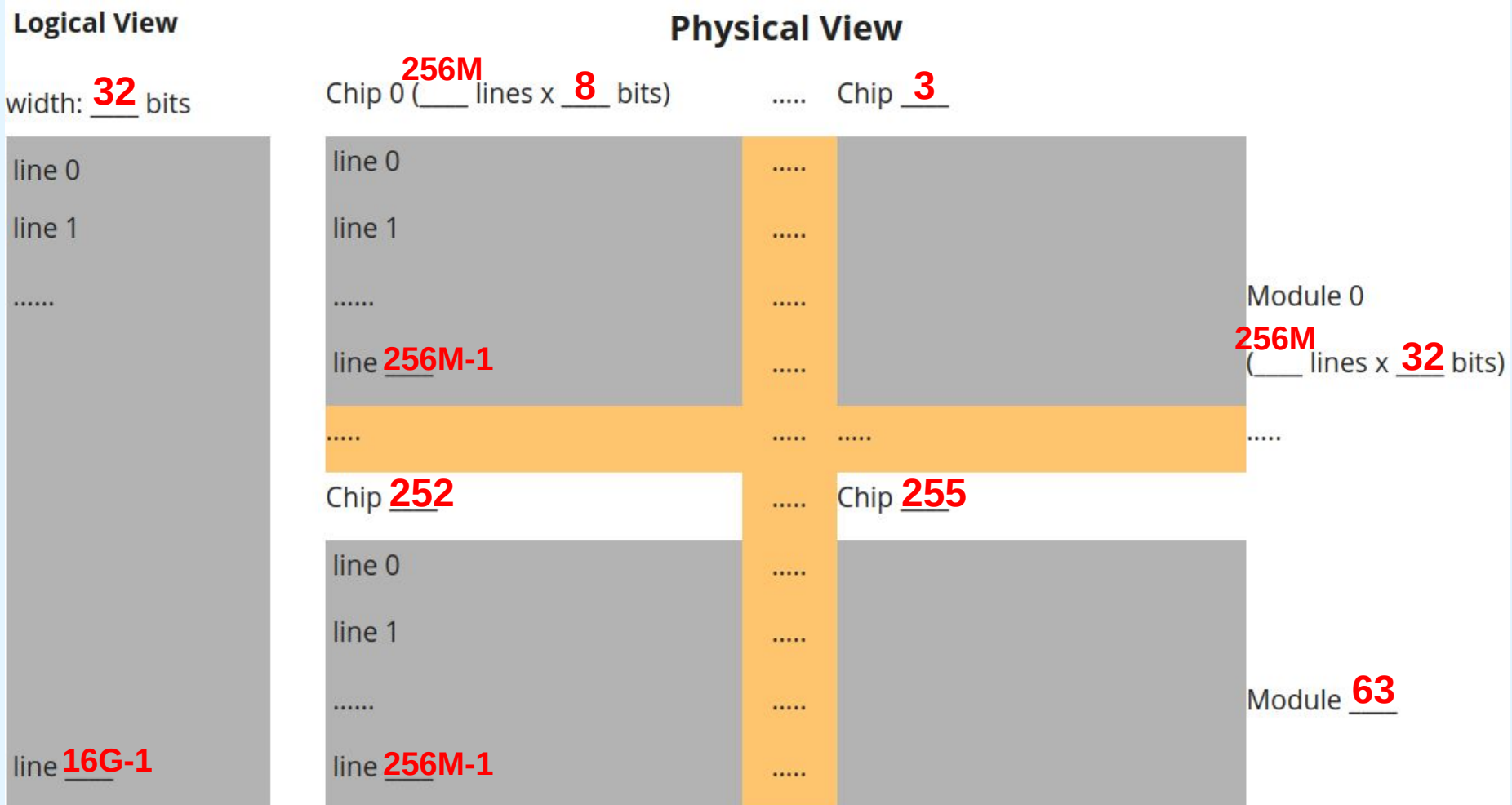
MP8 A memory of 16G x 32 bits is built with 256M x 8 chips and is word-addressable (32 bits word size). Answer the following questions.

g) Complete the representation bellow of the logical & physical view of memory:



MP8 A memory of 16G x 32 bits is built with 256M x 8 chips and is word-addressable (32 bits word size). Answer the following questions.

g) Complete the representation bellow of the logical & physical view of memory:
Solution:



MP8 A memory of 16G x 32 bits is built with 256M x 8 chips and is word-addressable (32 bits word size). Answer the following questions.

h) Find and present the module and the offset (line) within that module, for the memory address 8888888, considering the following alternatives for the mapping of a memory address into its physical components:

h1) High-Order Interleaving:

module: {?}

offset: {?}

h2) Low-Order Interleaving:

module: {?}

offset: {?}

MP8 A memory of 16G x 32 bits is built with 256M x 8 chips and is word-addressable (32 bits word size). Answer the following questions.

h) Find and present the module and the offset (line) within that module, for the memory address 8888888, considering the following alternatives for the mapping of a memory address into its physical components:

Solution:

h1) High-Order Interleaving:

module: {0}

offset: {8888888}

h2) Low-Order Interleaving:

module: {56}

offset: {138888}

MP8 A memory of 16G x 32 bits is built with 256M x 8 chips and is word-addressable (32 bits word size). Answer the following questions.

i) Provide the overall storage capacity (in bytes) of the memory.

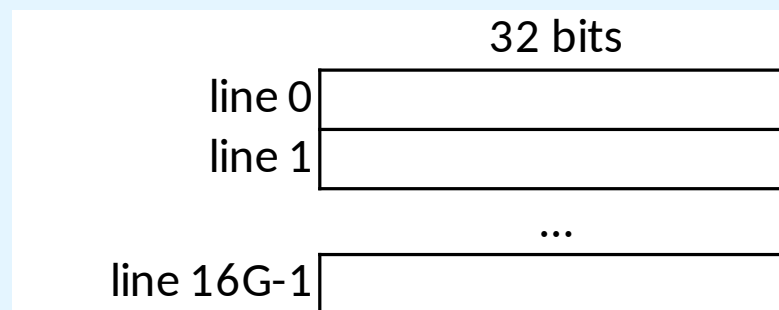
{?} bytes

MP8 A memory of 16G x 32 bits is built with 256M x 8 chips and is word-addressable (32 bits word size). Answer the following questions.

i) Provide the overall storage capacity (in bytes) of the memory.

Solution:

{**64G**} bytes



$$\begin{aligned}\text{\#BYTES(memory)} &= \text{\#LINES(memory)} \times \text{\#BYTES (line)} \\ &= 16\text{G} \times (32 / 8) \\ &= 16\text{G} \times 4 \\ &= 64\text{G bytes}\end{aligned}$$

MP9 Given a memory of 2048 bytes consisting of several 64 Byte x 8 chips, and assuming byte-addressable memory, which of the following seven alternatives uses the address bits correctly ?

- a.

2 bits for chip select	8 bits for address on chip
------------------------	----------------------------
- b.

16 bits for chip select	48 bits for address on chip
-------------------------	-----------------------------
- c.

6 bits for chip select	5 bits for address on chip
------------------------	----------------------------
- d.

1 bit for chip select	5 bits for address on chip
-----------------------	----------------------------
- e.

5 bits for chip select	6 bits for address on chip
------------------------	----------------------------
- f.

4 bits for chip select	6 bits for address on chip
------------------------	----------------------------
- g.

8 bits for chip select	56 bits for address on chip
------------------------	-----------------------------

MP9 Given a memory of 2048 bytes consisting of several 64 Byte x 8 chips, and assuming byte-addressable memory, which of the following seven alternatives uses the address bits correctly ?

- a.

2 bits for chip select	8 bits for address on chip
------------------------	----------------------------

 ← 10-bit address →
- b.

16 bits for chip select	48 bits for address on chip
-------------------------	-----------------------------

 ← 64-bit address →
- c.

6 bits for chip select	5 bits for address on chip
------------------------	----------------------------

 ← 11-bit address →
- d.

1 bit for chip select	5 bits for address on chip
-----------------------	----------------------------

 ← 6-bit address →
- e.

5 bits for chip select	6 bits for address on chip
------------------------	----------------------------

 ← 11-bit address →
- f.

4 bits for chip select	6 bits for address on chip
------------------------	----------------------------

 ← 10-bit address →
- g.

8 bits for chip select	56 bits for address on chip
------------------------	-----------------------------

 ← 64-bit address →

Solution:

CACHE1 A computer has a main memory of 2^{20} words and a direct mapped cache of 32 blocks, where each cache block contains 16 words.

- a) How many blocks of memory are there ?**
- b) What is the format of a memory address as seen by the cache ?**
- c) To which cache block will the memory address $0DB63_{16}$ map ?**
- d) To which memory block and word will the memory address $0DB63_{16}$ map ?**

Note: in this and in the following exercises on caches, “memory” refers to the “main memory”, also sometimes referred to as “RAM”.

CACHE1

A computer has a main memory of 2^{20} words and a direct mapped cache of 32 blocks, where each cache block contains 16 words.

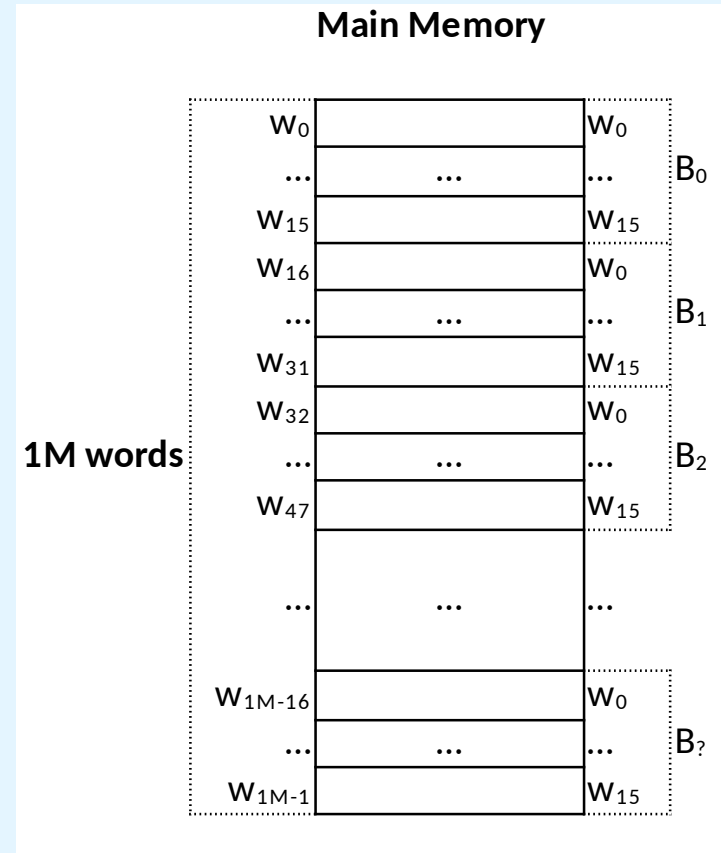
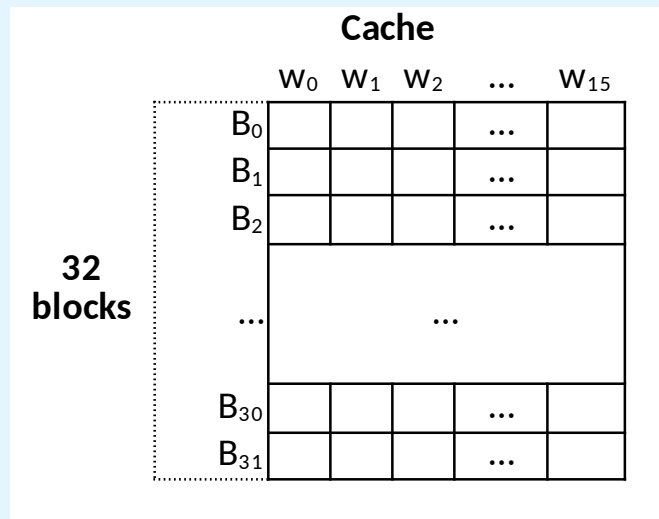
Based on the data given, one can build the following representations:

Problem Data:

#WORDS(memory) = 1M = 2^{20} words

#BLOCKS(cache) = 32 = 2^5 blocks

#WORDS(cache block) = 16 = 2^4 words



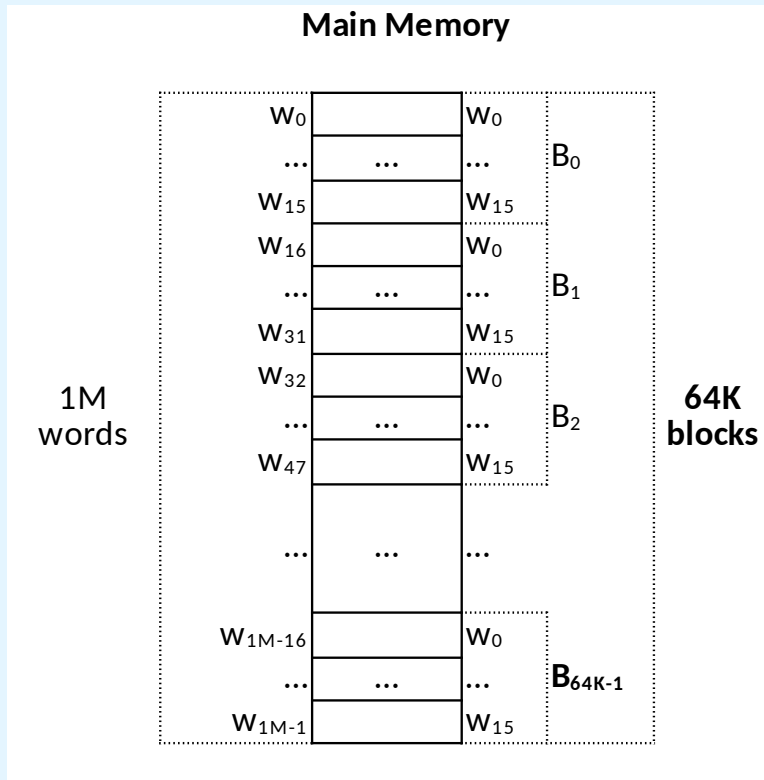
CACHE1 A computer has a main memory of 2^{20} words and a direct mapped cache of 32 blocks, where each cache block contains 16 words.

a) How many blocks of memory are there?

Solution:

#WORDS(memory block)
= #WORDS(cache block)
= 16 words

#BLOCKS(memory)
= #WORDS(memory) /
#WORDS(memory block)
= $2^{20} / 2^4 = 2^{16}$
= 65536 = **64K blocks**



CACHE1 A computer has a main memory of 2^{20} words and a direct mapped cache of 32 blocks, where each cache block contains 16 words.

b) What is the format of a memory address as seen by the cache ?

Solution:

- cache view: **memory address**=<tag, block, word>
- **#BITS(memory address)** = $\log_2(\text{\#WORDS}(\text{memory})) = \log_2(2^{20}) = 20 \text{ bits}$
- **#BITS(word)** = $\log_2(\text{\#WORDS}(\text{cache block})) = \log_2(16) = 4 \text{ bits}$
- **#BITS(block)** = $\log_2(\text{\#BLOCKS}(\text{cache})) = \log_2(32) = 5 \text{ bits}$
- **#BITS(tag)** = $\text{\#BITS}(\text{memory address}) - (\text{\#BITS}(\text{word}) + \text{\#BITS}(\text{block}))$
= $20 - (4 + 5)$
= **11 bits**

CACHE1 A computer has a main memory of 2^{20} words and a direct mapped cache of 32 blocks, where each cache block contains 16 words.

b) What is the format of a memory address as seen by the cache ?

Solution (cont.):

Additional notes:

20 bits		
tag	block	word
11 bits	5 bits	4 bits
2048 groups, of 32 blocks each	32 blocks, of 16 words each	16 words

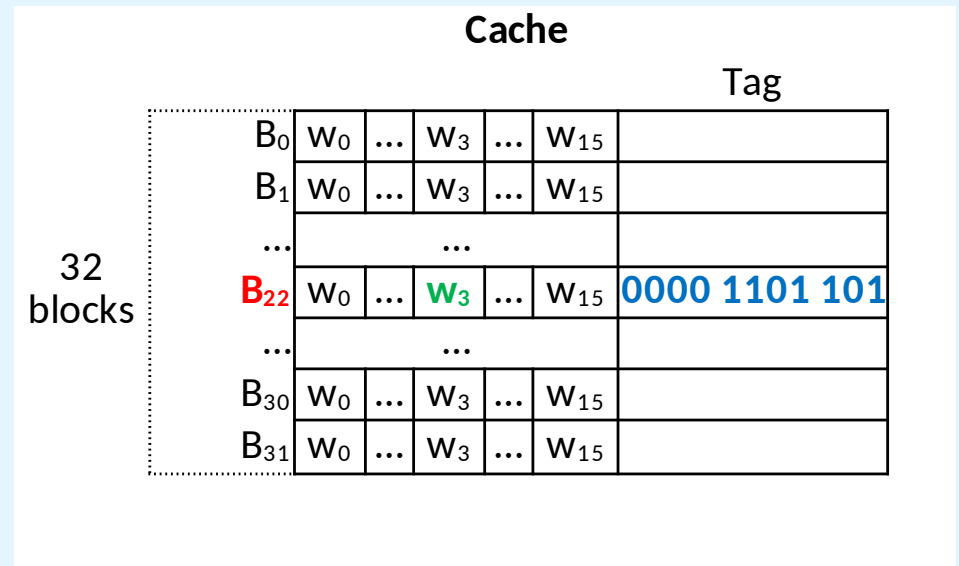
- the mapping between memory blocks and cache blocks repeats every **32** memory blocks (**block** = **block memory** % **32**) and such happens **2048** times
- thus, memory blocks **0,1,...,31** are mapped to the cache blocks **0,1,...,31**; memory blocks **32,33,...,63** are mapped to cache blocks **0,1,...,31**; and so on ...
- the **tag** field of a cache block states which memory block was copied to that cache block, from **2048** possible candidates

CACHE1 A computer has a main memory of 2^{20} words and a direct mapped cache of 32 blocks, where each cache block contains 16 words.

c) To which cache block will the memory address $0DB63_{16}$ map?

Solution: $0DB63_{16} = 0000\ 1101\ 1011\ 0110\ 0011_2$

20 bits		
tag	block	word
11 bits	5 bits	4 bits
0000 1101 101	1 0110	0011
(tag 109)	(block 22)	(word 3)



CACHE1 A computer has a main memory of 2^{20} words and a direct mapped cache of 32 blocks, where each cache block contains 16 words.

d) To which memory block and word will the memory address $0DB63_{16}$ map?

Solution:

- cache view:

$$0DB63_{16} = 0000\ 1101\ 1011\ 0110\ 0011_2 = 56163_{10}$$

- memory view:

memory address = <memory block, word>

- the **memory block** is found by joining the **tag** and **block** bits of the address from the cache view, that is:

$$\text{memory block} = \underline{000\ 1101\ 1011}\ 0110_2 = 3510_{10}$$

- the **word** component is given by the remaining bits:

$$\text{word} = 0011_2 = 3$$

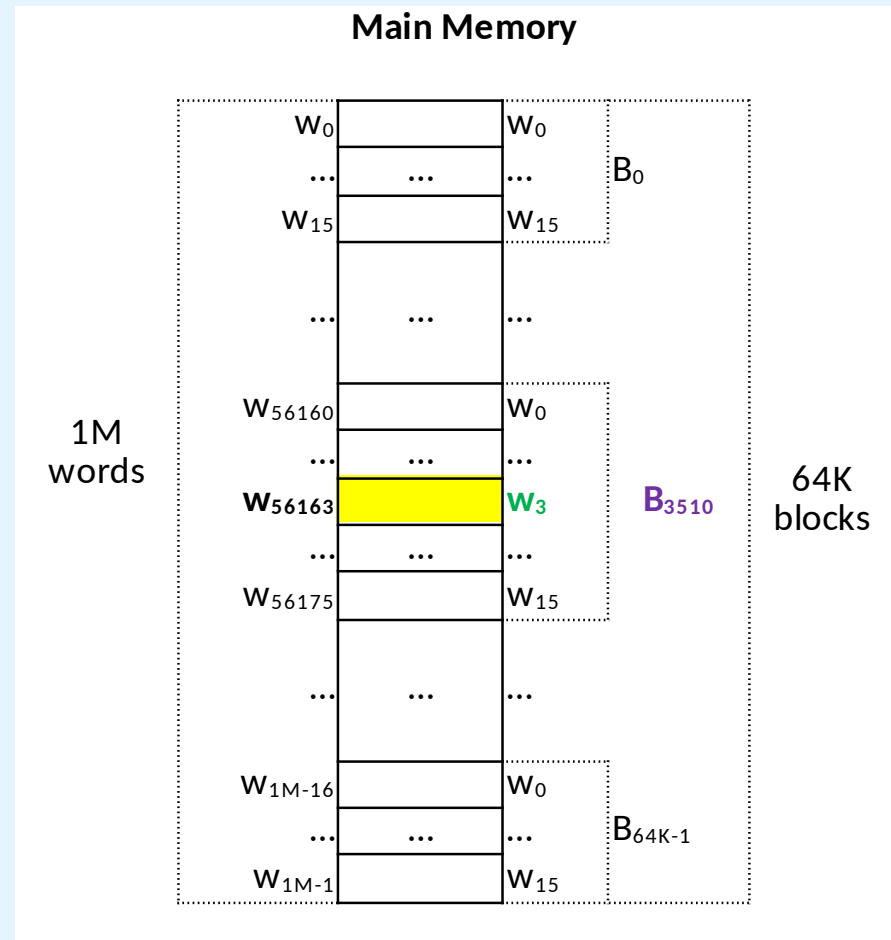
CACHE1 A computer has a main memory of 2^{20} words and a direct mapped cache of 32 blocks, where each cache block contains 16 words.

d) To which memory block and word will the memory address $0DB63_{16}$ map?

Solution (cont.):

Notice:

- $56160 = 3510 \times 16$
- $56163 = 3510 \times 16 + 3$



CACHE2 A computer has a main memory of 4G words and a direct mapped cache of 1K blocks, where each cache block contains 32 words.

- a) How many blocks of memory are there?**
- b) What is the format of a memory address as seen by the cache ?**
- c) To which cache block the memory address $000063FA_{16}$ maps ?**
- d) To which memory block, and block word, the memory reference $000063FA_{16}$ maps ?**

Homework

CACHE3/EAT4 A computer has a main memory of 4K blocks (of 8 words per block) and a direct-mapped cache of 8 blocks.

- a) What is the format of a memory address as seen by the cache ?**
- b) Compute the hit ratio for a program that loops 2 times from locations 0 to 67 in memory. The cache is initially empty.**
- c) Assume access time for the cache is 22ns and the time required to fill a cache slot from main memory is 300ns. Compute the effective access time in the b) conditions.**

CACHE3/EAT4 A computer has a main memory of 4K blocks (of 8 words per block) and a direct-mapped cache of 8 blocks.

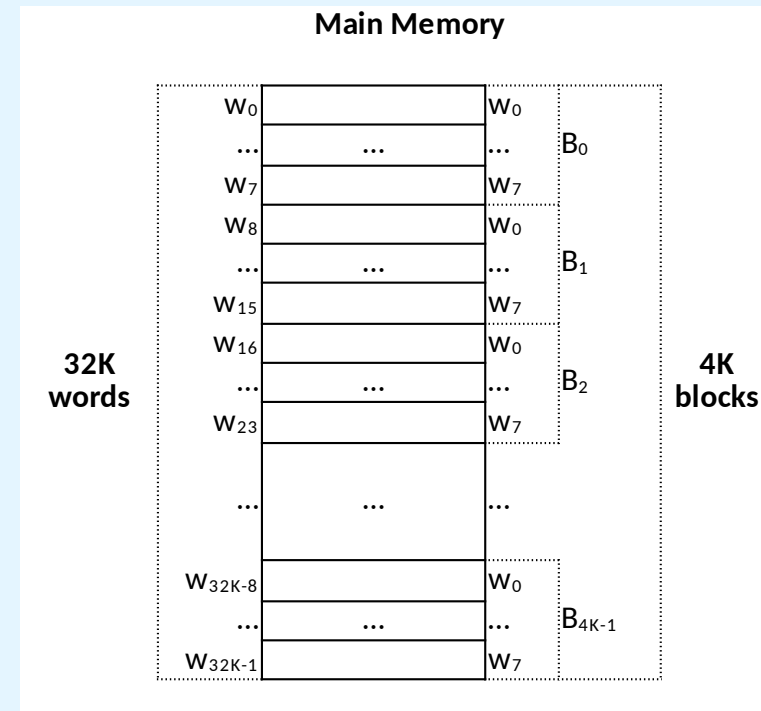
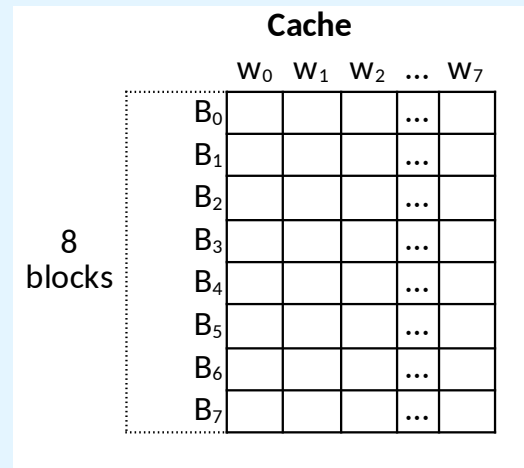
Based on the data given, one can build the following representations:

Problem Data:

#BLOCKS(memory) = 4K blocks = $2^2 \times 2^{10} = 2^{12}$ blocks

#WORDS(memory block) = 8 = 2^3 words

#BLOCKS(cache) = 8 blocks = 2^3 blocks



Consequences:

#WORDS(cache block) = #WORDS(memory block) = 8 words

#WORDS(memory) = #BLOCKS(memory) x #WORDS(memory block)

= $2^{12} \times 2^3 = 2^{15} = 32K$ words

CACHE3/EAT4 A computer has a main memory of 4K blocks (of 8 words per block) and a direct-mapped cache of 8 blocks.

a) What is the format of a memory address as seen by the cache ?

Solution:

15 bits		
tag	block	word
9 bits	3 bits	3 bits
512 groups, of 8 blocks each	8 blocks, of 8 words each	8 words

- cache view: **memory address**=<tag, block, word>
- $\#BITS(\text{memory address}) = \log_2(\#WORDS(\text{memory})) = \log_2(2^{15}) = 15 \text{ bits}$
- $\#BITS(\text{word}) = \log_2(\#WORDS(\text{cache block})) = \log_2(2^3) = 3 \text{ bits}$
- $\#BITS(\text{block}) = \log_2(\#BLOCKS(\text{cache})) = \log_2(2^3) = 3 \text{ bits}$
- $\#BITS(\text{tag}) = \#BITS(\text{memory address}) - (\#BITS(\text{word}) + \#BITS(\text{block}))$
 $= 15 - (3 + 3) = 9 \text{ bits}$

CACHE3/EAT4 A computer has a main memory of 4K blocks (of 8 words per block) and a direct-mapped cache of 8 blocks.

b) Compute the hit ratio for a program that loops 2 times from locations 0 to 67 in memory. The cache is initially empty.

Solution:

1st round of the loop

- access to memory word 0 implies a cache MISS and the loading of memory block 0 (with memory words 0 to 7) to the cache block 0
- access to memory words 1 to 7 implies 7 cache HITS
- access to memory words 8 to 63, from memory blocks 1 to 7, has similar implications: 1 MISS and 7 HITS per each block, for a total of 7 MISSES and 49 HITS; cache blocks 1 to 7 get copies of memory blocks 1 to 7
- access to memory word 64 implies a cache MISS, once its hosting memory block (8) also maps to cache block 0 but this cache block holds a copy of memory block 0; memory block 8 (with words 64 to 71) is then copied to cache block 0
- access to memory words 65 to 67 implies 3 HITS
- **at the end of loop 1: #MISSES = 1 + 7 + 1 = 9; #HITS = 7 + 49 + 3 = 59**

CACHE3/EAT4 A computer has a main memory of 4K blocks (of 8 words per block) and a direct-mapped cache of 8 blocks.

b) Compute the hit ratio for a program that loops 2 times from locations 0 to 67 in memory. The cache is initially empty.

Solution (cont.):

2nd round

- access to memory word 0 implies a cache MISS, once its hosting memory block (0) also maps to cache block 0 but this cache block holds a copy of memory block 8; memory block 0 (with words 0 to 7) is then copied to cache block 0
- access to memory words 1 to 7 implies 7 HITS
- access to memory words 8 to 63, from memory blocks 1 to 7, implies $7 \times 8 = 56$ HITS, once cache blocks 1 to 7 hold copies of those memory blocks
- access to memory word 64 implies a cache MISS, once its hosting memory block (8) also maps to cache block 0 but this cache block holds a copy of memory block 0; memory block 8 (with words 64 to 71) is then copied to cache block 0
- access to memory words 65 to 67 implies 3 HITS
- **at the end of loop 2: #MISSES = 1 + 1 = 2; #HITS = 7 + 56 + 3 = 66**

CACHE3/EAT4 A computer has a main memory of 4K blocks (of 8 words per block) and a direct-mapped cache of 8 blocks.

- b) Compute the hit ratio for a program that loops 2 times from locations 0 to 67 in memory. The cache is initially empty.**

Solution (cont.):

$$\text{\#ACCESSES} = 2 \times (67 - 0 + 1) = 2 \times 68 = 136$$

$$\text{\#MISSES} = 9 + 2 = 11$$

$$\text{\#HITS} = 59 + 66 = 125$$

$$\text{\textbf{HIT-RATIO}} = \text{\#HITS} / \text{\#ACCESSES} = 125 / 136 = \textbf{91,91\%}$$

CACHE3/EAT4 A computer has a main memory of 4K blocks (of 8 words per block) and a direct-mapped cache of 8 blocks.

- c) Assume access time for the cache is 22ns and the time required to fill a cache slot from main memory is 300ns. Compute the effective access time in the b) conditions.**

Solution:

- cache = level 1 of the hierarchy ; memory = level 2 of the hierarchy
- cache hit-rate = $H_1 = 0,9191$ (result of question b))
- cache access time = $T_1 = 22$ ns
- time to fill a cache slot (line) = $T_2 = 300$ ns
(time spent copying a block from memory into the cache)

overlapped access:

$$\begin{aligned} \text{EAT} &= H_1 \times T_1 + (1 - H_1) \times T_2 \\ &= 0,9191 \times 22 + (1 - 0,9191) \times 300 \\ &= 44,4902 \text{ ns} \end{aligned}$$

sequential access:

$$\begin{aligned} \text{EAT} &= H_1 \times T_1 + (1 - H_1) \times (T_1 + T_2) \\ &= 0,9191 \times 22 + (1 - 0,9191) \times (22 + 300) \\ &= 46,27 \text{ ns} \end{aligned}$$

CACHE4 A computer has a main memory of 2^{16} words and a fully associative cache of 64 blocks, with 32 words per cache block.

- a) How many blocks of memory are there?**
- b) What is the format of a memory address as seen by the cache ?**
- c) To which cache block will the memory address $F89C_{16}$ map ?**
- d) To which memory block, and block word, will the memory address $F89C_{16}$ map ?**

CACHE4 A computer has a main memory of 2^{16} words and a fully associative cache of 64 blocks, with 32 words per cache block.

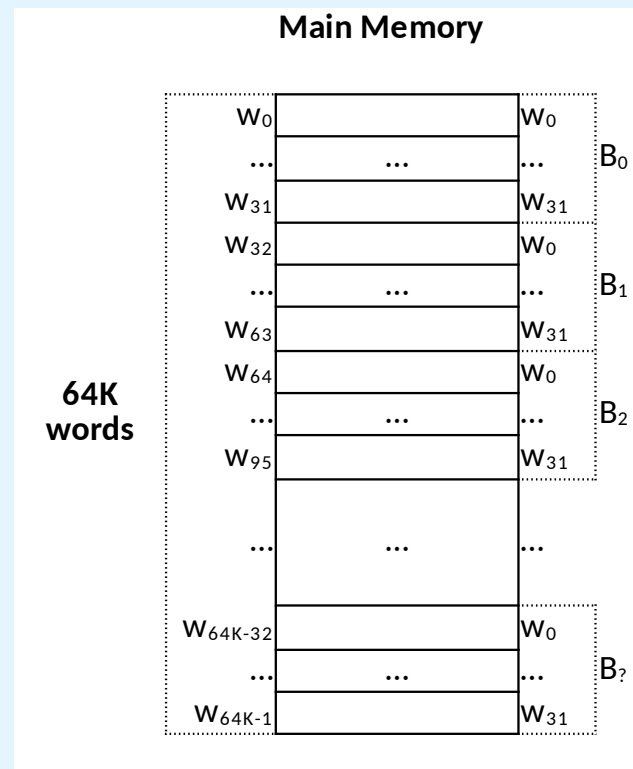
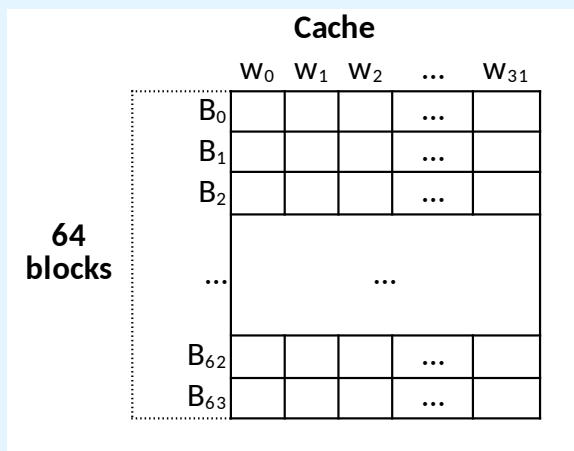
Based on the data given, one can build the following representations:

Problem Data:

#WORDS(memory) = 64K = 2^{16} words

#BLOCKS(cache) = 64 = 2^6 blocks

#WORDS(cache block) = 32 = 2^5 words



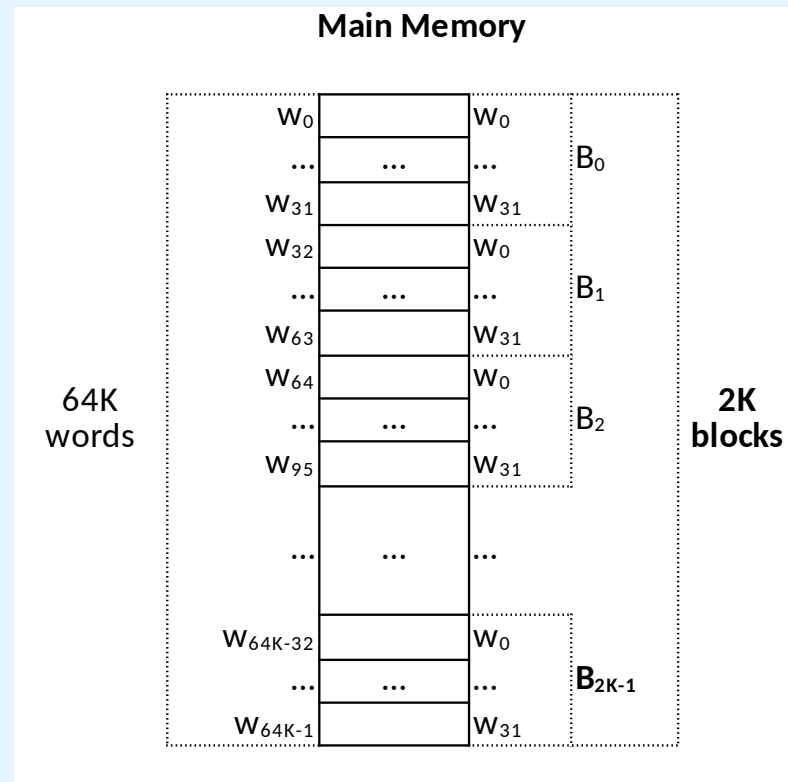
CACHE4 A computer has a main memory of 2^{16} words and a fully associative cache of 64 blocks, with 32 words per cache block.

a) How many blocks of memory are there?

Solution:

#WORDS(memory block)
= #WORDS(cache block)
= 32 words

#BLOCKS(memory)
= #WORDS(memory) /
#WORDS(memory block)
= $2^{16} / 2^5 = 2^{11}$
= 2048 = **2K blocks**



CACHE4 A computer has a main memory of 2^{16} words and a fully associative cache of 64 blocks, with 32 words per cache block.

b) What is the format of a memory address as seen by the cache ?

Solution:

- cache view: **memory address**=<**tag**, **word**>
- $\text{\#BITS}(\text{memory address}) = \log_2(\text{\#WORDS}(\text{memory})) = \log_2(2^{16}) = 16 \text{ bits}$
- $\text{\#BITS}(\text{word}) = \log_2(\text{\#WORDS}(\text{cache block})) = \log_2(2^5) = 5 \text{ bits}$
- $\text{\#BITS}(\text{tag}) = \text{\#BITS}(\text{memory address}) - \text{\#BITS}(\text{word})$
= $16 - 5$
= **11 bits**

CACHE4 A computer has a main memory of 2^{16} words and a fully associative cache of 64 blocks, with 32 words per cache block.

b) What is the format of a memory address as seen by the cache ?

Solution (cont.):

16 bits	
tag	word
11 bits	5 bits
2048 blocks, of 32 words each	32 words

Additional notes:

- the mapping between memory blocks and cache blocks is arbitrary:
 - a memory block may be copied to **any** cache block
- the **tag** field of a cache block states which memory block was copied to that cache block, from **2048** possible candidates

CACHE4 A computer has a main memory of 2^{16} words and a fully associative cache of 64 blocks, with 32 words per cache block.

c) To which cache block will the memory address $F89C_{16}$ map ?

Solution:

- Once the cache is associative, then any cache block may receive a copy of the memory block that contains the word having this address !
- Thus, without additional information about the current state of the cache, it is not possible to answer to this question.

CACHE4 A computer has a main memory of 2^{16} words and a fully associative cache of 64 blocks, with 32 words per cache block.

d) To which memory block, and block word, will the memory address $F89C_{16}$ map ?

Solution:

- cache view:

$$F89C_{16} = 1111\ 1000\ 1001\ 1100_2 = 63644_{10}$$

- memory view:

memory address = <memory block, word>

- the **memory block** is given by the bits of the **tag** of the address from the cache view, that is:

$$\text{memory block} = 1111\ 1000\ 100_2 = 1988_{10}$$

- the **word** component is given by the remaining bits:

$$\text{word} = 1\ 1100_2 = 28$$

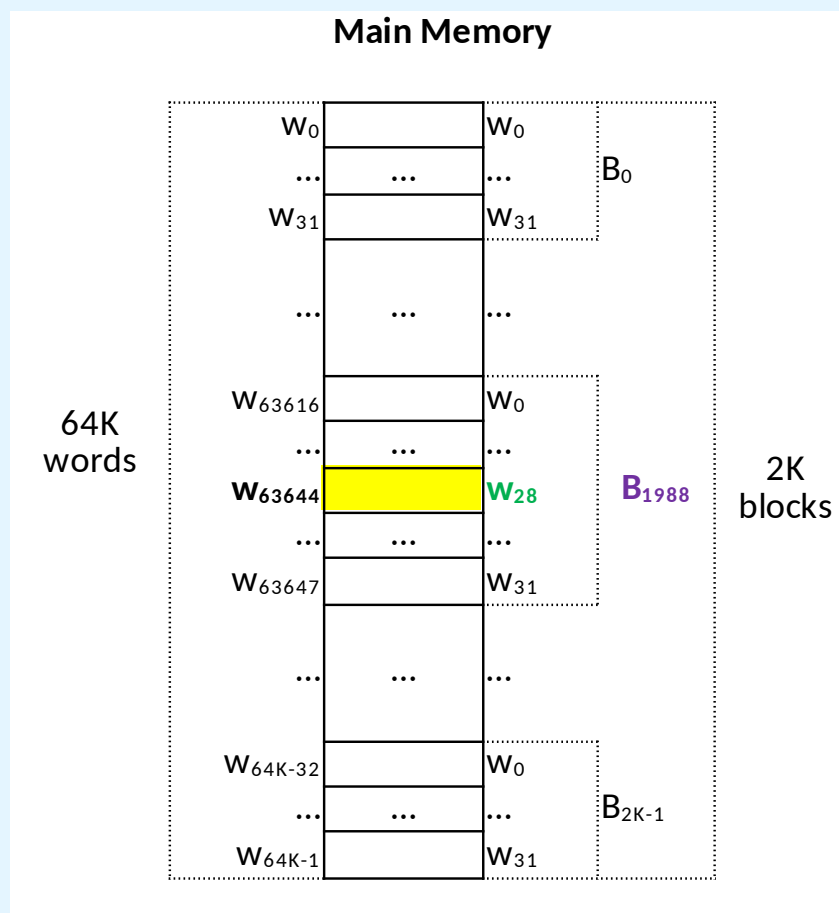
CACHE4 A computer has a main memory of 2^{16} words and a fully associative cache of 64 blocks, with 32 words per cache block.

d) To which memory block, and block word, will the memory address $F89C_{16}$ map ?

Solution (cont.):

Notice:

- $63616 = 1988 \times 32$
- $63644 = 1988 \times 32 + 28$



CACHE5 A computer has a main memory of 16M words and a fully associative cache of 128 blocks, with 64 words per cache block.

- a) How many blocks of memory are there?**
- b) What is the format of a memory address as seen by the cache ?**
- c) To which cache block will the memory address $01D872_{16}$ map ?**
- d) To which memory block, and block word, will the memory address $01D872_{16}$ map ?**

Homework

CACHE6 A computer has a main memory of 128M words and a 2-way set associative cache of 32K blocks, with 64 words per cache block.

- a) How many blocks of memory are there?**
- b) What is the format of a memory address as seen by the cache ?**
- c) To which cache set, and block of that set, will the memory address $0DB630D_{16}$ map?**
- d) To which memory block, and block word, will the memory address $0DB630D_{16}$ map ?**

CACHE6 A computer has a main memory of 128M words and a 2-way set associative cache of 32K blocks, with 64 words per cache block.

Based on the data given, one can build the following representations:

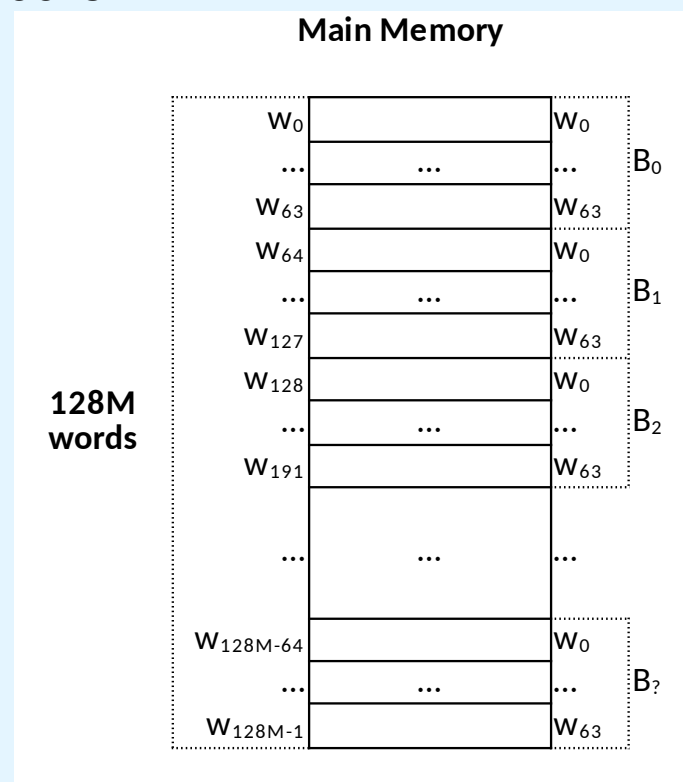
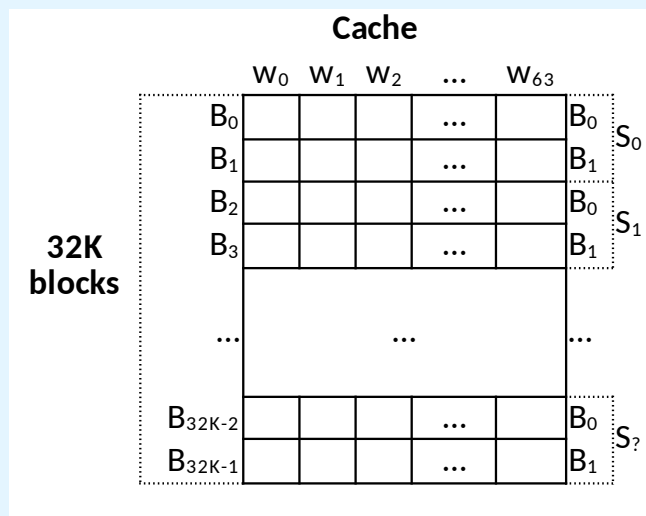
Problem Data:

#WORDS (memory) = 128M = $2^7 \times 2^{20} = 2^{27}$ words

N = number of ways = number of cache blocks per cache set = 2

#BLOCKS(cache) = 32K = $2^5 \times 2^{10} = 2^{15}$ blocks

#WORDS(cache block) = 64 = 2^6 words



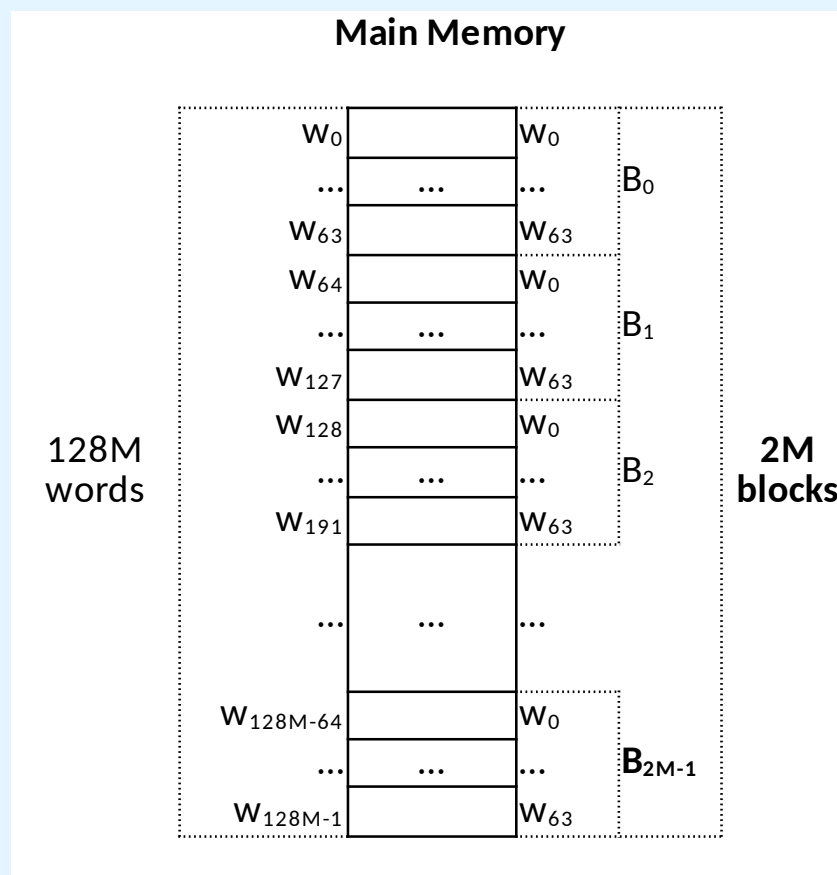
CACHE6 A computer has a main memory of 128M words and a 2-way set associative cache of 32K blocks, with 64 words per cache block.

a) How many blocks of memory are there?

Solution:

$$\begin{aligned}\text{\#WORDS(memory block)} &= \text{\#WORDS(cache block)} \\ &= 64 \text{ words}\end{aligned}$$

$$\begin{aligned}\text{\#BLOCKS(memory)} &= \text{\#WORDS(memory)} / \\ &\quad \text{\#WORDS(memory block)} \\ &= 2^{27} / 2^6 = 2^{21} \\ &= 2^2 \times 2^{20} = \mathbf{2M \text{ blocks}}\end{aligned}$$



CACHE6 A computer has a main memory of 128M words and a 2-way set associative cache of 32K blocks, with 64 words per cache block.

b) What is the format of a memory address as seen by the cache ?

Solution:

- cache view: **memory address**=<tag, set, word>
- $\#BITS(\text{memory address}) = \log_2(\#WORDS(\text{memory})) = \log_2(2^{27}) = 27 \text{ bits}$
- $\#BITS(\text{word}) = \log_2(\#WORDS(\text{cache block})) = \log_2(2^6) = 6 \text{ bits}$
- $\#SETS(\text{cache}) = \#BLOCKS(\text{cache}) / N = 2^{15} / 2^1 = 2^{14}$
- $\#BITS(\text{set}) = \log_2(\#SETS(\text{cache})) = \log_2(2^{14}) = 14 \text{ bits}$
- $\#BITS(\text{tag}) = \#BITS(\text{memory address}) - (\#BITS(\text{word}) + \#BITS(\text{set}))$
 $= 27 - (6 + 14) = 7 \text{ bits}$

CACHE6 A computer has a main memory of 128M words and a 2-way set associative cache of 32K blocks, with 64 words per cache block.

b) What is the format of a memory address as seen by the cache ?

Solution (cont.):

27 bits		
tag	set	word
7 bits	14 bits	6 bits
128 candidate blocks, for each set	16K sets (of N=2 blocks each)	64 words

Additional notes:

- a memory block will be copied always to the **same cache set** (that is, memory blocks are directly mapped to a certain cache set)
- a memory block will be copied to **any of the N=2 blocks** of its **cache set** (and in that set, the search for a block is done associatively)
- the **tag** field of a cache block identifies which candidate memory block was copied to that cache block, from **128** possible memory blocks

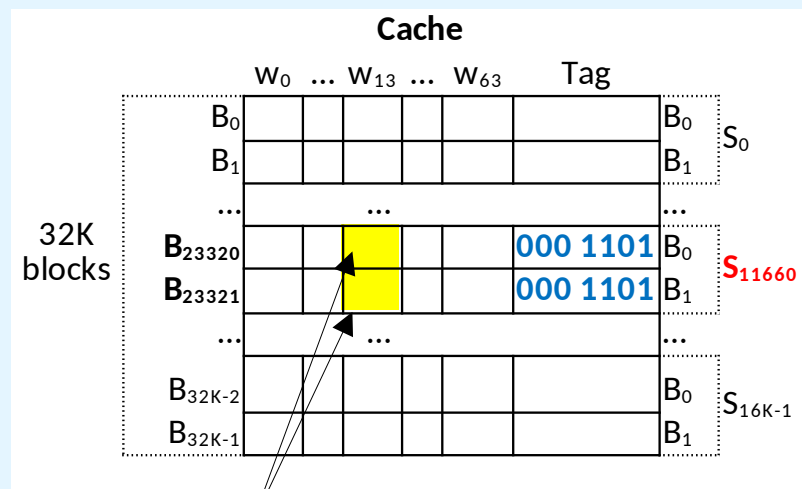
CACHE6 A computer has a main memory of 128M words and a 2-way set associative cache of 32K blocks, with 64 words per cache block.

c) To which cache set, and block of that set, will the memory address $0DB630D_{16}$ map?

Solution: $0DB630D_{16} =$

$0000\ 1101\ 1011\ 0110\ 0011\ 0000\ 1101_2$

the bit 0 comes from the conversion of the leftmost hexadecimal digit (0), and does not belong to the address, once this one has only 27 bits, and not 28



the block of the set may be **any**, once the cache is set-associative

20 bits

tag	set	word
7 bits	14 bits	6 bits
000 1101	1011 0110 0011 00	00 1101

(set 11660)

(word 13)

CACHE6 A computer has a main memory of 128M words and a 2-way set associative cache of 32K blocks, with 64 words per cache block.

d) To which memory block, and block word, will the memory address $0DB630D_{16}$ map ?

Solution:

- cache view:

$$0DB630D_{16} = 0000\ 1101\ 1011\ 0110\ 0011\ 0000\ 1101_2 = 14377741_{10}$$

- memory view:

memory address = <memory block, word>

- the **memory block** is found by joining the **tag** and **set** bits of the address from the cache view, that is:

$$\text{memory block} = \underline{0000\ 1101\ 1011\ 0110\ 0011\ 00}_2 = 224652_{10}$$

- the **word** component is given by the remaining bits:

$$\text{word} = 00\ 1101_2 = 13_{10}$$

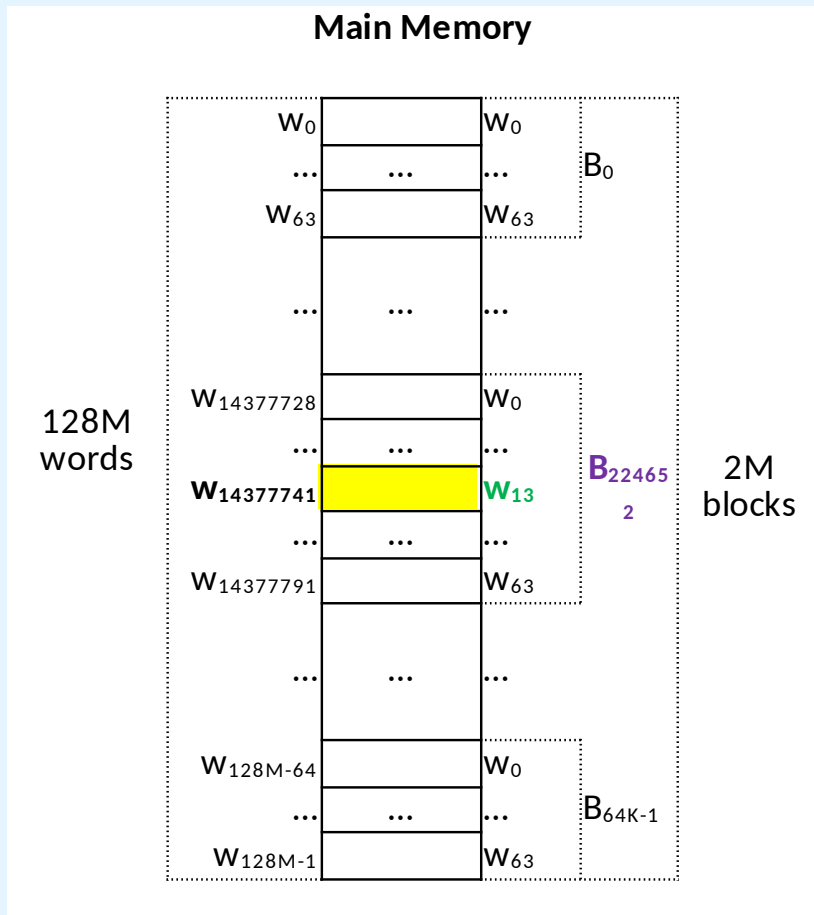
CACHE6 A computer has a main memory of 128M words and a 2-way set associative cache of 32K blocks, with 64 words per cache block.

d) To which memory block, and block word, will the memory address $0DB630D_{16}$ map ?

Solution (cont.):

Notice:

- $14377728 = 224652 \times 64$
- $14377741 = 224652 \times 64 + 13$



CACHE7 A computer has a byte-addressable memory with 24-bit addresses, and a 64KB cache with 32 bytes blocks.

Show the format of a memory address as seen by the cache if the cache is:

- a) direct mapped**
- b) (fully) associative**
- c) 4-way set associative**

Homework

CACHE8/EAT5 A computer has a main memory of 2K blocks, of 8 words each, and a 2-way set associative cache, with 4 sets of blocks.

- a) What is the format of a memory address as seen by the cache ?**
- b) Compute the hit ratio for a program that loops 3 times from locations 8 to 23, and 40 to 47, in memory.**

CACHE8/EAT5 A computer has a main memory of 2K blocks, of 8 words each, and a 2-way set associative cache, with 4 sets of blocks.

Based on the data given, one can build the following representations:

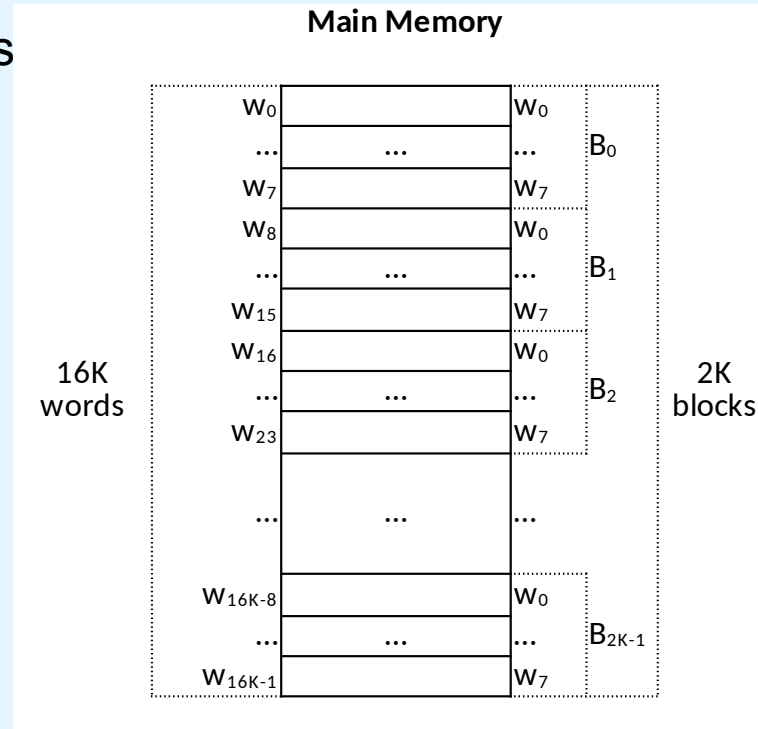
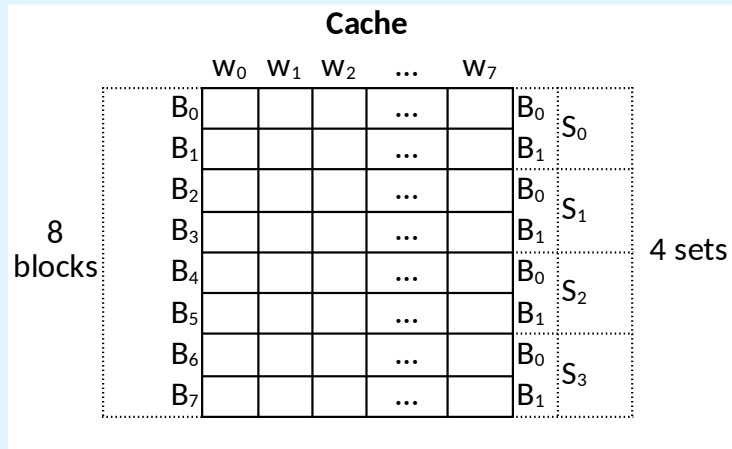
Problem Data:

#BLOCKS(memory) = $2K = 2^1 \times 2^{10} = 2^{11}$ blocks

#WORDS(memory block) = $8 = 2^3$ words

N = num. of ways = num. of blocks per set = 2

#SETS(cache) = $4 = 2^2$ sets



Consequences:

#BLOCKS(cache) = #SETS(cache) $\times$ N = $4 \times 2 = 8 = 2^3$ blocks

#WORDS(memory) = #BLOCKS(memory) $\times$ #WORDS(memory block)

= $2^{11} \times 2^3 = 2^{14} = 16K$ words

CACHE8/EAT5

A computer has a main memory of 2K blocks, of 8 words each, and a 2-way set associative cache, with 4 sets of blocks.

a) What is the format of a memory address as seen by the cache ?

Solution:

14 bits		
tag	set	word
9 bits	2 bits	3 bits
512 candidate blocks, for each set	4 sets (of N=2 blocks each)	8 words

- cache view: **memory address**=<tag, set, word>
- $\#BITS(\text{memory address}) = \log_2(\#WORDS(\text{memory})) = \log_2(2^{14}) = 14 \text{ bits}$
- $\#BITS(\text{word}) = \log_2(\#WORDS(\text{cache block})) = \log_2(2^3) = 3 \text{ bits}$
- $\#BITS(\text{set}) = \log_2(\#SETS(\text{cache})) = \log_2(2^2) = 2 \text{ bits}$
- $\#BITS(\text{tag}) = \#BITS(\text{memory address}) - (\#BITS(\text{word}) + \#BITS(\text{set}))$
 $= 14 - (3 + 2) = 9 \text{ bits}$

CACHE8/EAT5 A computer has a main memory of 2K blocks, of 8 words each, and a 2-way set associative cache, with 4 sets of blocks.

b) Compute the hit ratio for a program that loops 3 times from locations 8 to 23, and 40 to 47, in memory.

Solution:

1st round:

	W ₀	W ₁	W ₂	W ₃	W ₄	W ₅	W ₆	W ₇	Tag		
B ₀										B ₀	S ₀
B ₁										B ₁	
B ₂										B ₀	S ₁
B ₃										B ₁	
B ₄										B ₀	S ₂
B ₅										B ₁	
B ₆										B ₀	S ₃
B ₇										B ₁	



	W ₀	W ₁	W ₂	W ₃	W ₄	W ₅	W ₆	W ₇	Tag		
B ₀										B ₀	S ₀
B ₁										B ₁	
B ₂	W ₈	W ₉	W ₁₀	W ₁₁	W ₁₂	W ₁₃	W ₁₄	W ₁₅	000 000 000	B ₀	S ₁
B ₃										B ₁	
B ₄										B ₀	S ₂
B ₅										B ₁	
B ₆										B ₀	S ₃
B ₇										B ₁	

$8_{10} = 000000000 \text{ 01 } 000_2 \Rightarrow \text{tag}=0, \text{set}=1, \text{word}=0 \Rightarrow \text{MISS and block}=0$

$9_{10} = 000000000 \text{ 01 } 001_2 \Rightarrow \text{tag}=0, \text{set}=1, \text{word}=1 \Rightarrow \text{HIT in block 0}$

...

$15_{10} = 000000000 \text{ 01 } 111_2 \Rightarrow \text{tag}=0, \text{set}=1, \text{word}=7 \Rightarrow \text{HIT in block 0}$

**7
HITS**

CACHE8/EAT5

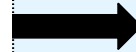
A computer has a main memory of 2K blocks, of 8 words each, and a 2-way set associative cache, with 4 sets of blocks.

b) Compute the hit ratio for a program that loops 3 times from locations 8 to 23, and 40 to 47, in memory.

Solution:

1st round
(cont.):

	W ₀	W ₁	W ₂	W ₃	W ₄	W ₅	W ₆	W ₇	Tag		
B ₀										B ₀	S ₀
B ₁										B ₁	
B ₂	W ₈	W ₉	W ₁₀	W ₁₁	W ₁₂	W ₁₃	W ₁₄	W ₁₅	000 000 000	B ₀	S ₁
B ₃										B ₁	
B ₄										B ₀	S ₂
B ₅										B ₁	
B ₆										B ₀	S ₃
B ₇										B ₁	



	W ₀	W ₁	W ₂	W ₃	W ₄	W ₅	W ₆	W ₇	Tag		
B ₀										B ₀	S ₀
B ₁										B ₁	
B ₂	W ₈	W ₉	W ₁₀	W ₁₁	W ₁₂	W ₁₃	W ₁₄	W ₁₅	000 000 000	B ₀	S ₁
B ₃										B ₁	
B ₄	W ₁₆	W ₁₇	W ₁₈	W ₁₉	W ₂₀	W ₂₁	W ₂₂	W ₂₃	000 000 000	B ₀	S ₂
B ₅										B ₁	
B ₆										B ₀	S ₃
B ₇										B ₁	

$16_{10} = 000000000 10 000_2 \Rightarrow \text{tag}=0, \text{set}=2, \text{word}=0 \Rightarrow \text{MISS and block}=0$

$17_{10} = 000000000 10 001_2 \Rightarrow \text{tag}=0, \text{set}=2, \text{word}=1 \Rightarrow \text{HIT in block 0}$

...

$23_{10} = 000000000 10 111_2 \Rightarrow \text{tag}=0, \text{set}=2, \text{word}=7 \Rightarrow \text{HIT in block 0}$

7
HITS

CACHE8/EAT5

A computer has a main memory of 2K blocks, of 8 words each, and a 2-way set associative cache, with 4 sets of blocks.

b) Compute the hit ratio for a program that loops 3 times from locations 8 to 23, and 40 to 47, in memory.

Solution:

1st round
(cont.):

	W ₀	W ₁	W ₂	W ₃	W ₄	W ₅	W ₆	W ₇	Tag		
B ₀										B ₀	S ₀
B ₁										B ₁	
B ₂	W ₈	W ₉	W ₁₀	W ₁₁	W ₁₂	W ₁₃	W ₁₄	W ₁₅	000 000 000	B ₀	S ₁
B ₃										B ₁	
B ₄	W ₁₆	W ₁₇	W ₁₈	W ₁₉	W ₂₀	W ₂₁	W ₂₂	W ₂₃	000 000 000	B ₀	S ₂
B ₅										B ₁	
B ₆										B ₀	S ₃
B ₇										B ₁	



	W ₀	W ₁	W ₂	W ₃	W ₄	W ₅	W ₆	W ₇	Tag		
B ₀										B ₀	S ₀
B ₁										B ₁	
B ₂	W ₈	W ₉	W ₁₀	W ₁₁	W ₁₂	W ₁₃	W ₁₄	W ₁₅	000 000 000	B ₀	S ₁
B ₃	W ₄₀	W ₄₁	W ₄₂	W ₄₃	W ₄₄	W ₄₅	W ₄₆	W ₄₇	000 000 001	B ₁	S ₁
B ₄	W ₁₆	W ₁₇	W ₁₈	W ₁₉	W ₂₀	W ₂₁	W ₂₂	W ₂₃	000 000 000	B ₀	S ₂
B ₅										B ₁	
B ₆										B ₀	S ₃
B ₇										B ₁	

$40_{10} = 000000001 \text{ } 01 \text{ } 000_2 \Rightarrow \text{tag}=1, \text{set}=1, \text{word}=0 \Rightarrow \text{MISS and block}=1$

$41_{10} = 000000001 \text{ } 01 \text{ } 001_2 \Rightarrow \text{tag}=1, \text{set}=1, \text{word}=1 \Rightarrow \text{HIT in block 1}$

...

$47_{10} = 000000001 \text{ } 01 \text{ } 111_2 \Rightarrow \text{tag}=0, \text{set}=1, \text{word}=7 \Rightarrow \text{HIT in block 1}$

7
HITS

CACHE8/EAT5 A computer has a main memory of 2K blocks, of 8 words each, and a 2-way set associative cache, with 4 sets of blocks.

b) Compute the hit ratio for a program that loops 3 times from locations 8 to 23, and 40 to 47, in memory.

Solution:

2nd and 3rd rounds:

	W ₀	W ₁	W ₂	W ₃	W ₄	W ₅	W ₆	W ₇	Tag		
B ₀										B ₀	S ₀
B ₁										B ₁	
B ₂	W ₈	W ₉	W ₁₀	W ₁₁	W ₁₂	W ₁₃	W ₁₄	W ₁₅	000 000 000	B ₀	S ₁
B ₃	W ₄₀	W ₄₁	W ₄₂	W ₄₃	W ₄₄	W ₄₅	W ₄₆	W ₄₇	000 000 001	B ₁	
B ₄	W ₁₆	W ₁₇	W ₁₈	W ₁₉	W ₂₀	W ₂₁	W ₂₂	W ₂₃	000 000 000	B ₀	S ₂
B ₅										B ₁	
B ₆										B ₀	S ₃
B ₇										B ₁	

$8_{10} = 000000000 \text{ 01 } 000_2 \Rightarrow \text{tag}=0, \text{set}=1, \text{word}=0 \Rightarrow \text{HIT in block 0}$

...

$15_{10} = 000000000 \text{ 01 } 111_2 \Rightarrow \text{tag}=0, \text{set}=1, \text{word}=7 \Rightarrow \text{HIT in block 0}$

$16_{10} = 000000000 \text{ 10 } 000_2 \Rightarrow \text{tag}=0, \text{set}=2, \text{word}=0 \Rightarrow \text{HIT in block 0}$

...

$23_{10} = 000000000 \text{ 10 } 111_2 \Rightarrow \text{tag}=0, \text{set}=2, \text{word}=7 \Rightarrow \text{HIT in block 0}$

$40_{10} = 000000001 \text{ 01 } 000_2 \Rightarrow \text{tag}=1, \text{set}=1, \text{word}=0 \Rightarrow \text{HIT in block 1}$

...

$47_{10} = 000000001 \text{ 01 } 111_2 \Rightarrow \text{tag}=1, \text{set}=1, \text{word}=7 \Rightarrow \text{HIT in block 1}$

8
HITS

8
HITS

8
HITS

CACHE8/EAT5 A computer has a main memory of 2K blocks, of 8 words each, and a 2-way set associative cache, with 4 sets of blocks.

- b) Compute the hit ratio for a program that loops 3 times from locations 8 to 23, and 40 to 47, in memory.**

Solution: (cont.):

$$\text{\#ACCESSES} = 3 \times [(23-8+1) + (47-40+1)] = 3 \times 24 = 72$$

$$\text{\#MISSES} = 3$$

$$\text{\#HITS} = (7+7+7) + 2 \times (8+8+8) = 21 + 48 = 69$$

$$\text{HIT-RATIO} = \text{\#HITS} / \text{\#ACCESSES} = 69 / 72 = \mathbf{95,83\%}$$